

Mizar Evo

Detailed discussion deck with presenter notes for the Białystok Mizar team

Mizar Evo project

September 2026

1.1 - Three Pillars (1/2)

The design is judged by three constraints that must hold at the same time.

Pillar	What it means	Why it matters
Readability	Proof scripts stay close to mathematical exposition; local disambiguation is preferred when ambiguity matters.	Human reviewers can inspect the argument without replaying hidden automation.
AI-readiness	Verifier context, source spans, obligations, and safe edit classes are exposed through small auditable interfaces.	AI can help with search, explanation, migration, and patch proposals without becoming trusted proof authority.
Scalability	Imports, artifacts, packages, caches, and publication links have stable boundaries.	Large-library maintenance becomes predictable without changing theorem meaning.

The design is judged by three constraints that must hold at the same time.

File	Global names	Global names
Readability	Print scripts may show in mathematical notation; local names like <code>alpha</code> is preferred when available in scripts.	There are requirements for the readability of the script: replacing <code>alpha</code> with <code>alpha</code> is preferred.
Modularity	Global context, macro syntax, definitions, and variable names are exposed through an all-encompassing interface.	It can help with search, explanation, migration, and patch propagation. It can become a great tool for the user.
Scalability	Impacts, analysis, packages, and packages. It can be used to build a new system.	Large-scale maintenance is possible without changing the meaning.

Presenter script: Read aloud:

- The design is judged by three constraints that must hold at the same time.

For the table, read the row labels first, then the design reason.

The three pillars reinforce each other:

Readable source

- > better human review
- > better AI retrieval and edits
- > clearer artifacts and dependency traces
- > more reliable large-library maintenance

Design test:

- A feature is good only if it improves automation or scale while preserving readable mathematics.

The three pillars reinforce each other:

Readable source
→ better human review
→ better AS verification and edits
→ cleaner artifacts and dependency traces
→ more reliable large-library maintenance

Design time:

• A feature is good only if it improves automation or scale while preserving readable mathematics.

Presenter script: Read aloud:

- A feature is good only if it improves automation or scale while preserving readable mathematics.

After the example, say which design obligation the syntax makes visible.

Key Phrase:

Mizar Evo should not make proofs shorter by making the argument invisible.

Presenter script: Say:

- This is the key contrast with very tactic-heavy workflows.
- Some automation is valuable, but the accepted proof artifact must remain inspectable.

Key Points:

- A first-order set-theoretic core matches much ordinary mathematical practice.
- ATP tools are strong in first-order reasoning.
- Soft types, modes, attributes, and registrations can remain Mizar-facing structure around that core.
- The kernel can focus on checking evidence rather than performing high-level elaboration.

Design Reason:

- This choice preserves the current Mizar foundation while letting powerful ATPs work at the layer where they are strongest.
- It also avoids making dependent-type elaboration or tactic execution part of the trusted explanation of ordinary mathematical text.

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Key Points:

- explicit module imports;
- templates for parameterized definitions and theorem schemas;
- package manifests and lockfiles;
- stable fully-qualified names;
- deterministic build artifacts;

Key Points:

- incremental and parallel verification;
- LSP for editors;
- planned MCP-style context surfaces for AI agents;
- versioned links from publications to library objects.

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- LSP for editors;
- shared MC-Rule content engine for all agents;
- versioned links from packages to library objects.

Presenter script: Read aloud:

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Key Points:

- not abandoning declarative proof style;
- not making ATP success trusted by default;
- not replacing human mathematical review with AI;
- not forcing every old article into a new style at once.

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- Make dependencies and names more explicit.
- Separate structure definition, inheritance mapping, and template parameterization.
- Give tools stable identities and source spans.
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2.2 - Mizar To Evo Language Delta Inventory (1/11)

This is a checklist, not a claim that every item must be finalized before the visit. Artifact and workflow deltas that affect migration semantics follow in the next inventory slide.

Area	Current Mizar baseline	Evo delta	Why it changes
Article environment	<code>environ</code> roles such as vocabularies, notations, constructors, registrations, requirements	<code>explicit import</code> prelude and module/package namespace	make the dependency surface reproducible and reviewable
Module interfaces	article export is mediated by library tooling, with fewer source-level interface boundaries	<code>export</code> , <code>public/private</code> , opaque imports, and separate compilation define the public API surface	keep reusable interfaces stable while proof bodies and helpers stay private

Part 2. Language Specification

2.2 - Mizar To Evo Language Delta Inventory (1/11)

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Area	Current <i>libar</i> baseline	Goal state	Why it changes
Article order in module	non-lexical rules such as alphabetical, natural, chronological, registration, registration etc.	lexical: super package and sub package name space	make the dependency pattern repeatable and testable
Module interface	article type is not addressed by library tools, with former source-level interface boundaries	key part, package/package, package in point, and package can publish data in the public API	keep module interface stable with the goal of better and better may publish

Presenter script: Say:

- The important review question is not "how many syntactic changes exist?" but "which Mizar-facing idea is being preserved, and which hidden dependency is being made explicit?"

Read aloud:

- This is a checklist, not a claim that every item must be finalized before the visit. Artifact and workflow deltas that affect migration semantics follow in the next inventory slide.

For the table, read the row labels first, then the design reason.

2.2 - Mizar To Evo Language Delta Inventory (2/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Symbol identity	article-local names and MML article labels	path-derived FQNs and module-qualified labels	support migration, renaming, API compatibility, and citation repair
Symbolic notation and aliases	broad symbolic constructor vocabulary, with synonyms and antonyms as mathematical aliases	arbitrary symbols concentrated in <code>func</code> and <code>pred</code> ; type constructors stay identifier-like; synonym/antonym equivalence is preserved	stabilize lexing, parsing, editor tooling, and AI edits without losing alternative notation

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (2/11)

Area	Current Mizar handles	Evo takes	What it changes
Symbol identity	arbitrary symbol name and MML article to do	pre-defined ID for named and qualified labels	support migration, renaming, APL from parity, and ID change
Symbolic variables and aliases	bound symbols, constructor variables, both synonym and antonym as mathematical objects	arbitrary symbols concentrated in fun and pred type constructors; synonym/antonym equivalence is presumed	handles binding, parsing, pointer testing, and all definitions having alternative notations

Presenter script: For the table, read the row labels first, then the design reason. Use this as the transition: Mizar To Evo Language Delta Inventory (2/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (3/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Operator precedence and parse domains	notation is powerful but precedence choices can be implicit in parser/verifier behavior	explicit operator metadata, separate term/formula precedence domains, and parse-before-overload discipline	make grouping deterministic, reviewable, and independent of overload choice
Lexical activation	vocabulary and notation availability is article-environment driven	import-prelude pre-scan, source-position active lexicon, no forward references, operator metadata after declaration	make tokenization and dot/operator disambiguation deterministic

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (3/11)

Area	Current Mizar handles	Evo adds	What it changes
Operator precedence and parsing rules	operator precedence is determined by the order of operators in the expression	operator precedence is determined by the order of operators in the expression, and the order of operators in the expression is determined by the order of operators in the expression	operator precedence is determined by the order of operators in the expression, and the order of operators in the expression is determined by the order of operators in the expression
Local variables	variables are declared at the beginning of a block and are only visible within that block	variables are declared at the beginning of a block and are only visible within that block, and variables are declared at the beginning of a block and are only visible within that block	variables are declared at the beginning of a block and are only visible within that block, and variables are declared at the beginning of a block and are only visible within that block

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Use this as the transition: Mizar To Evo Language Delta Inventory (3/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (4/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Soft type foundation	Mizar soft types over set-theoretic objects	preserve soft typing while making radix/mode heads, object/set, type erasure, widening/narrowing, and reconsider obligations explicit	keep Mizar's foundation recognizable while exposing verifier obligations
Structures	parent structure, fields, and selectors are tightly coupled in one declaration	struct, field, property, and inherit are separate concepts	distinguish layout, derived canonical values, and inherited views

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (4/11)

2.2 - Mizar To Evo Language Delta Inventory (4/11)

Area	Current: Mizar handles	Evo adds	Why it changes
Sort type foundation	Mizar sort types are non-constructive objects	Constructive sort typing with making sorts/modes heads, as just sort type forms, $\text{sort}(m)/\text{sort}(n)$, and $\text{sort}(m) \leq \text{sort}(n)$ relations	Evo's type checker recognizes sorts recognizing sort m relations
Structures	parent structure, fields, and subtypes are tightly coupled in one declaration	Structs, Enums, properties, and subtypes are separate concepts	Distinguish layers, desired canonicalization, and intended uses

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Mizar To Evo Language Delta Inventory (4/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (5/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Inheritance	inherited fields are mostly implicit in structure syntax	<code>inherit Child</code> extends <code>Parent</code> where ... records mapping, renaming, narrowing, and coherence	make multiple inheritance and diamond cases auditable
Modes and attributes	soft type vocabulary with adjectives and clusters	modes remain type abbreviations/refinements; attributes are type-refining predicates	preserve Mizar style while separating classification from data layout

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (5/11)

Area	Current Mizar handles	Evo does	What it changes
Intelligence	Intelligence takes an explicit input in interactive system	Intelligent Child receives external source ... receives mapping, reasoning, learning, and conclusion	Intelligence is more multiple behavior on and then and comes as a whole
Modes and attributes	Multi-type vocabulary with adjectives and classes	Multi-type vocabulary with adjectives/relationships/attributes are representing prediction	grammar: Mizar style multi-lingualizing characteristics from data format

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Mizar To Evo Language Delta Inventory (5/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (6/11)

Area	Current Mizar baseline	Evo delta	Why it changes
<code>sethood</code> and comprehensions	Fraenkel-style set formation relies on set-theoretic side conditions	mode <code>sethood</code> and comprehension checks are explicit type-checker obligations	preserve set-theoretic foundations before ATP search starts
Definitions and correctness	<code>func/pred</code> definitions use familiar <code>equals/means</code> styles	preserve <code>equals</code> , <code>means</code> , <code>existence</code> , <code>uniqueness</code> , and <code>assume</code> side conditions as explicit obligations	keep Mizar's definitional idiom while making obligations auditable

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└ 2.2 - Mizar To Evo Language Delta Inventory (6/11)

Area	Current Mizar handles	Evo delta	What it changes
prelude and comprehension	Formal logic notation and comprehension conditions	Modern syntax and comprehension checks are simplified type-checker obligations	premise mechanism: use datums before AT P search is in
Definitions and theorems	Formal logic notation and axioms/lemmas	premise agents, names, new syntax, new operators, and various other conditions as explicit obligations	Every Mizar's definitional datum is also using obligations as explicit

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Mizar To Evo Language Delta Inventory (6/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (7/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Predicate and functor properties	algebraic properties support concise automatic reasoning	proof-backed property declarations such as commutativity, symmetry, and irreflexivity	keep compact mathematical notation while recording exactly what automation may use
Overload and redefine	overloaded symbols and redefinitions are central Mizar idioms	root/family overload resolution, coherence with, and refinement joins are recorded and explainable	avoid source-order ambiguity and make refined result facts stable

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (7/11)

Area	Current Mizar header	Evo delta	Why it changes
Procedures and function definitions	Algorithmic properties support: can do parametric reasoning	Pre-stANDARD properties: declarations such as commutative, idempotent, and associative	Keep on top of mathematical notation. Not recording exactly what is written may cause
Context and evaluation	overload of symbols and notations are central Mizar ideas	To support the overload symbols, constants, and values are recorded and verifiable	avoid notation-related ambiguity and make notation more readable

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Mizar To Evo Language Delta Inventory (7/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (8/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Registrations, clusters, and reductions	powerful automatic propagation and simplification, often hard to explain locally	labeled registration items, import-filtered resolution graph, reduction index, trace artifacts	keep automation but make inference and rewriting explainable
Schemes and generics	classical <code>scheme</code> , <code>of/over</code> parameterization	first-class templates; bracket form is canonical, constraints and inference are explicit, <code>of/over</code> are shorthands	unify parameterized definitions and theorem schemas without hiding instantiation choices

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (8/11)

Area	Current Mizar headers	Evo delta	What it changes
Declarations, lemmas, and reductions	potential automatic propagation and simplification, often hard to explain locally	explicit registration items, in part -> mod reduction graph, reduction items, item attributes	deep automation handles inference and providing explicit ask
Schemas and generic	logical schema, schema param specification	first-class template blocks, form is canonical, constraints and invariants are explicit, schema are distributed	early parametric definitions and schema schemas with a hiding schema hidden choices

Presenter script: For the table, read the row labels first, then the design reason. Use this as the transition: Mizar To Evo Language Delta Inventory (8/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (9/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Declarative proof skeleton	Jaśkowski-style proof ... end text with let, assume, thus, hence, thesis, and diffuse reasoning	preserve the readable proof skeleton and emit extracted obligations, thesis states, and source spans as metadata	protect Mizar's proof prose while making proof state toolable
Proof status	verified theorem items as the normal publication unit	open, assumed, and conditional theorem statuses	separate unsettled work, assumptions, and clean verified results

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (9/11)

Area	Current Mizar header	Evo delta	What it changes
Declaration prelude	... now (not) with let, use case, case, hence, because, and define because by	promote the readable prelude before and omit extraneous subgoals, lambda terms, and macros appear as in Mizar	promote Mizar's prelude while making prelude readable
Proof status	part of theorem from as hypothesis and justification	types, axioms, and contradictions: theorem status	separate as needed mark, assumptions, and show part of proofs

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Mizar To Evo Language Delta Inventory (9/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (10/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Proof citations	by references to visible facts	grouped/bulk citations, used-axiom recording, citation refinement	support small verifier-checked repairs and AI-assisted edits
Type views	existing qua, implicit widening, and local disambiguation idioms	keep source-level qua, and emit resolved view/coercion metadata artifacts	make overload and inherited-view choices inspectable

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (10/11)

Area	Current Mizar symbols	Evo delta	Why it changes
Preconditions	by assumption is valid fact	assumable choices, ass-schem reasoning, choices statement	support multi well-checked repairs and skolemized rules
Type items	ending, open, explicit ending, and local demonstrations items	open multi-level ques and end-symbols item / exercise in a data definition	to use variables and behavioral choices programmable

Presenter script: For the table, read the row labels first, then the design reason.
Use this as the transition: Mizar To Evo Language Delta Inventory (10/11). Point to the visible example, question, or diagram before moving on.

2.2 - Mizar To Evo Language Delta Inventory (11/11)

Area	Current Mizar baseline	Evo delta	Why it changes
Algorithms	mathematical proof language, not a general verified programming surface	<code>algorithm</code> , contracts, invariants, termination, MVM, by <code>computation</code>	support verified computation without changing theorem truth

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└ Part 2. Language Specification

└ 2.2 - Mizar To Evo Language Delta Inventory (11/11)

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Mizar To Evo Language Delta Inventory (11/11). Point to the visible example, question, or diagram before moving on.

Area	Current Mizar handles	Evo takes	Why it changes
Algorithms	mathematical proof language, not a general purpose programming language	algorithms, contracts, formalisms, in particular, MVVM, by composition	support unified computational models changing internal state

2.3 - Mizar To Evo Artifact And Workflow Delta Inventory (1/4)

These rows are not purely syntactic, but they affect what migration tools, reviewers, editors, and packages can trust.

Area	Current Mizar baseline	Evo delta	Why it changes
Annotations	comments and informal tool hints	semantic-neutral annotations such as <code>@proof_hint</code> , <code>@show_type</code> , <code>@show_resolution</code> , <code>@latex</code>	guide tools while keeping logical meaning unchanged
Diagnostics and LSP	verifier messages are primarily human-facing output	stable diagnostic codes, primary/secondary spans, fix suggestions, lazy explanations, and LSP records	make failures actionable for humans, editors, and AI tools

Mizar Evo

└ Part 2. Language Specification

└ 2.3 - Mizar To Evo Artifact And Workflow Delta Inventory
(1/4)

These rows are not purely syntactic, but they affect what migration tools, reviewers, editors, and packages can trust.

Area	Current Mizar handles	Evo delta	Why it changes
Annotations	comments and internal is defined	comments are not treated as type of delta, delta_type, delta_value, delta value	delta_type is logical meaning or change
Diagonals and 15.9	delta_type is not a delta_type output	delta_type is not a delta_type output	delta_type is not a delta_type output

Presenter script: Say:

- These are artifact deltas, but they belong near the language discussion because they determine whether a migrated source change remains explainable.

Read aloud:

- These rows are not purely syntactic, but they affect what migration tools,
- reviewers, editors, and packages can trust.

For the table, read the row labels first, then the design reason.

2.3 - Mizar To Evo Artifact And Workflow Delta Inventory (2/4)

Area	Current Mizar baseline	Evo delta	Why it changes
ATP boundary	proof search and verifier behavior are not surfaced as a stable artifact boundary	ATP dispatch plus independently checked certificates and rejection categories	keep a small trusted base
Packages and dependency resolution	article/MML workflow	mizar.pkg, lockfile, SemVer, features, compatibility checks, cached verifier artifacts	enable reproducible, package-oriented development

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└ Part 2. Language Specification

└ 2.3 - Mizar To Evo Artifact And Workflow Delta Inventory
(2/4)

2.3 - Mizar To Evo Artifact And Workflow Delta Inventory (2/4)

Area	Current Mizar workflow	Evo delta	Why it changes
ATP boundary	proof search and rejection behavior are not isolated as a single artifact boundary	ATP dispatch rules independently checked with call-in and rejection capabilities	keep a small trusted base
Packaging and dependency resolution	strictly MM1 workflow	mizar-jar, lockfile, 5m Ver, mizar, com-pubkey checks, cached earlier artifacts	enable crypto lockfile, package-oriented development

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Mizar To Evo Artifact And Workflow Delta Inventory (2/4). Point to the visible example, question, or diagram before moving on.

2.3 - Mizar To Evo Artifact And Workflow Delta Inventory (3/4)

Area	Current Mizar baseline	Evo delta	Why it changes
Incremental verification	accepted article output is the main reuse boundary	dependency slices, VC anchors, witness hashes, and cache keys; cache reuse is never proof authority	scale verification while preserving clean-build equivalence
Documentation generation	source comments and publication pages are separate from reusable API documentation	<code>mizar doc</code> , <code>::: doc</code> comments, label/FQN cross-links, <code>@latex</code> , docs from verified artifacts	produce browsable API documentation without making docs a proof gate

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└ Part 2. Language Specification

└ 2.3 - Mizar To Evo Artifact And Workflow Delta Inventory (3/4)

2.3 - Mizar To Evo Artifact And Workflow Delta Inventory (3/4)

Area	Current: Mizar handles	Evo delta	Why it changes
Recommendations	recommend article output to the main team boundary	depends on: docs, VC, anchors, m-team handbook, and cache key; cache team to move post-authority	code mitigation while preserving check-back equivalence
Documentation generation	generate comments and post-authority page on a separate from teambook API documentation	docs: v0, 1.1 doc team meta, build API N team-meta, docs doc team meta; docs from north of artifact	pre documentation with API doc team meta; docs team making docs a post gate

Presenter script: For the table, read the row labels first, then the design reason. Use this as the transition: Mizar To Evo Artifact And Workflow Delta Inventory (3/4). Point to the visible example, question, or diagram before moving on.

2.3 - Mizar To Evo Artifact And Workflow Delta Inventory (4/4)

Area	Current Mizar baseline	Evo delta	Why it changes
Code extraction	no general verified extraction workflow in the source language	terminating algorithms, ghost erasure, target-neutral runtime IR, extractor configuration	keep executable output downstream of verified computation
Formalized Mathematics link	article and library identity are closely coupled	publication metadata links to reusable library modules	preserve citation value while supporting package reuse

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└ Part 2. Language Specification

└ 2.3 - Mizar To Evo Artifact And Workflow Delta Inventory
(4/4)

Area	Current Mizar handles	Evo adds	Why it changes
Code extraction	no general method extraction method is in the source language	embedding algorithm, gho m stream, page-to-xml routines IR, abstract configuration	deep extractable output downstream of verified computation
Formalised Mathematics	proofs and theory theorems are already encoded	publications in mizar delta is research library in delta	promotes cleaner rules and the supporting package rules

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Mizar To Evo Artifact And Workflow Delta Inventory (4/4). Point to the visible example, question, or diagram before moving on.

Message:

- This chapter is the part we should not compress.
- The Bialystok review should treat the language surface as a specification, not only as a design story.
- Each syntax change should be reviewed for three things: readability for Mizar authors, migration cost for MML, and determinism for parsers, verifiers, and tools.
- The examples below are not final tutorial material. They are review targets: if a construct looks too unlike Mizar, or if a hidden Mizar idiom is missing, that is exactly the feedback we need.

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Canonical specification sources: All paths in this table are under `doc/spec/en/`.

Topic	Source
Cross-chapter grammar	<code>appendix_a.grammar_summary.md</code>
Operator precedence	<code>appendix_b.operator_precedence.md</code> , <code>10.functors.md</code>
Lexical structure	<code>02.lexical_structure.md</code>
Structures	<code>05.structures.md</code>
Attributes and modes	<code>06.attributes.md</code> , <code>07.modes.md</code>

└ Part 2. Language Specification

└ 2.5 - Grammar Sources For This Review (1/2)

Canonical specification sources: All paths in this table are under doc/spec/en/.	
Topic	0 index
Formal logical grammar	01 formal2_4_grammar_2_grammar_01.md
Operational grammar	01 formal2_4_operational_grammar_01.md
Logical notation	02 logical_notation_01.md
Logical notation	03 logical_notation_01.md
Notation	04 notation_01.md
All Notation and more	05 notation_01.md, 07 more_01.md

Presenter script: Read aloud:

- All paths in this table are under doc/spec/en/.

For the table, read the row labels first, then the design reason.

Slow down: this is language-specification review, not only implementation detail.

2.5 - Grammar Sources For This Review (2/2)

Topic	Source
Terms and formulas	13.term_expression.md, 14.formulas.md
Statements and proofs	15.statements.md, 16.theorems_and_proofs.md
Registrations and templates	17.clusters_and_registrations.md, 18.templates.md

Review rule:

- The slides summarize the surface grammar. The spec files remain the authority for edge cases and well-formedness rules.

└ Part 2. Language Specification

└ 2.5 - Grammar Sources For This Review (2/2)

Topic	Source
Terms and notations	01.terms_and_notations.mz, 01.terms.mz
Statements and proofs	02.statements.mz, 02.statements_and_proofs.mz
Definitions and axioms	03.definitions_and_axioms.mz, 03.definitions.mz

Review rule:

- The slides summarize the surface grammar. The spec files remain the authority for edge cases and well-formedness rules.

Presenter script: Read aloud:

- The slides summarize the surface grammar. The spec files remain the authority for edge cases and well-formedness rules.

For the table, read the row labels first, then the design reason.

Slow down: this is language-specification review, not only implementation detail.

The review should cover this grammar stack in order:

- 1 lexical classes and active lexicon;
- 2 module/import prelude and item activation;
- 3 definition blocks and visibility;
- 4 type expressions, attributes, modes, and structures;
- 5 functors, predicates, notation, properties, and overload families;

The review should cover this grammar stack in order:

- 1 operator precedence and term/formula boundary;
- 2 terms, formulas, comprehensions, the, sethood, and qua;
- 3 theorem/proof statements and citations;
- 4 registrations, reductions, templates, annotations, and algorithms.

The review should cover this grammar stack in order:

- lexical classes and active lexicon;
- module/import prelude and item activation;
- definition blocks and visibility;
- type expressions, attributes, modes, and structures;
- functors, predicates, notation, properties, and overload families;

The review should cover this grammar stack in order:

- operators: precedence and term/formula boundary;
- terms, formulas, comprehension, this, attribute, and equ;
- theorem/proof statements and lemmas;
- right rules, red axioms, lemmas, and notations, and algorithms.

Presenter script: Read aloud:

- lexical classes and active lexicon;
- module/import prelude and item activation;
- definition blocks and visibility;
- type expressions, attributes, modes, and structures;
- functors, predicates, notation, properties, and overload families;

Slow down: this is language-specification review, not only implementation detail.

Why this order matters:

- Mizar-style readability is not only local syntax. It depends on when names become visible, which notation is active, and which automation is allowed to fire.

Presenter script: Read aloud:

- Mizar-style readability is not only local syntax. It depends on when names become visible, which notation is active, and which automation is allowed to fire.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
compilation_unit ::= import_prelude export_prelude  
                  { annotated_declaration } ;  
  
import_prelude   ::= { import_stmt } ;  
export_prelude   ::= { export_stmt } ;  
  
declaration      ::= definition_block  
                  | reserve_decl  
                  | registration_block  
                  | claim_block
```

Grammar sketch:

```
                  | [ visibility ] theorem_item  
                  | [ visibility ] notation_decl ;  
  
visibility        ::= "private" | "public" ;
```



```

Grammar::block:
compilation_unit ::= * ( import_prelude export_prelude
                      { unnamed_declaration } ) ;

import_prelude  ::= { import_stmt } ;
export_prelude  ::= { export_stmt } ;

declaration ::= definition_block
             | reserve_decl
             | registration_block
             | extern_block

```

```

Grammar::block:
| | visibility | reserved_name
| | visibility | notation_decl ;

visibility ::= "private" | "public" ;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Compilation Unit Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Example:

```
import algebra.group as group;
export algebra.group;

public theorem left_cancel:
  for G being Group, a, b, c being Element of G
  st a * b = a * c holds b = c;
```

Review Point:

- The file begins with a reproducible dependency surface, then ordinary Mizar items. This replaces article-environment roles without turning the language into a generic programming module system.

Example:

```
import m3_group, group as group;
export m3_group, group;

public theorem left_cancell:
  for G being Group, m, b, c being Element of G
  ex a = b * m * c holds b = c;
```

Review Point:

- The file begins with a reproducible dependency surface, then ordinary Mizar items. This replaces article-environment roles without turning the language into a generic programming module system.

Presenter script: Read aloud:

- The file begins with a reproducible dependency surface, then ordinary Mizar items. This replaces article-environment roles without turning the language into a generic programming module system.

After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
import_stmt ::= "import" module_alias_decl  
              { ",", module_alias_decl } ";" ;  
export_stmt ::= "export" module_path { ",", module_path } ";" ;  
  
module_alias_decl ::= module_path [ "as" module_identifier ]  
                  | module_branch_import ;
```

Example:

```
import algebra.group as group;  
import algebra.ring.{Ring, Ideal};  
export algebra.group;
```

└ Part 2. Language Specification

└ 2.8 - Import Prelude And Item Activation (1/2)

Grammar sketch:

```
import_stmt ::= "import" module_name_decl  
            { ":", module_name_decl } ";" ;  
export_stmt ::= "export" module_name { ":", module_name } ";" ;  
module_name_decl ::= module_name [ "as" module_identifier ]  
                  module_name_import ;
```

Example:

```
import algebra.group as group;  
import algebra.ring_of_ring, idem;  
export algebra.group;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Import Prelude And Item Activation (1/2). Point to the visible example, question, or diagram before moving on.

Semantic rule:

- All `import` statements must appear before the first non-import item.
- Imported symbols seed the active lexicon for the body.
- Declarations in the current module become active only after their item is complete.
- A later import-shaped statement is a syntax error, not a dynamic environment update.

Reason:

- This makes lexing, parsing, and AI context extraction deterministic.

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- Declarations in the current module become active only after their item is complete.
- A later import-shaped statement is a syntax error, not a dynamic environment update.
- This makes lexing, parsing, and AI context extraction deterministic.

Grammar sketch:

```
identifier      ::= ( letter | "_" )  
                  { letter | digit | "_" | "'" } ;  
constructor_name ::= identifier | readable_constructor_name ;  
user_symbol     ::= symbol_char { symbol_char } ;
```

Example:

```
struct ZeroStr where  
  field carrier -> set;  
end;  
  
func + (x, y: Integer) -> Integer equals int.add(x, y);
```


└ Part 2. Language Specification

└ 2.9 - Lexical Classes (1/2)

Grammar sketch:

```
ident id_char ::= { letter | "_" }  
{ letter | digit | "_" } ** ;  
constructor_name ::= ident id_char | readable_constructor_name ;  
user_symbol ::= symbol_char { symbol_char } ;
```

Example:

```
axiom is true where  
  f is of carrier -> set;  
end;  
  
[func f (x, y: integer) => integer equals int.add(x, y);
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Lexical Classes (1/2). Point to the visible example, question, or diagram before moving on.

Policy:

- `mode`, `struct`, and `attr` names use constructor names.
- `field` and `property` names are identifiers.
- Arbitrary symbolic notation is concentrated in `func` and `pred`.
- Tokenization uses the longest active match at the source position.

Review Point:

- This is one of the largest visible syntax differences from legacy Mizar; it should be reviewed with real algebraic and set-theoretic examples.

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- Tokenization uses the longest active match at the source position.
- This is one of the largest visible syntax differences from legacy Mizar; it should be reviewed with real algebraic and set-theoretic examples.

Reserved punctuation includes:

```
, . .. ; : := ( ) [ ] { } .{ = <> & -> . = .* @[ ...
```

Dot policy:

- . can be selector syntax, namespace qualification, or a user functor symbol, depending on grammar position and the active lexicon.
- The parser preserves enough surface form for the resolver to classify it.

Bracket policy:

- [and] are reserved for template arguments and built-in bracket functor syntax.
- Postfix indexing such as `x[y]` is not silently admitted.

2.10 - Reserved Symbols And Dot Disambiguation (1/2)

a Postfix indexing such as `x[y]` is not silently admitted.

After the example, say which design obligation the syntax makes visible.

Example:

```
let R be Ring;  
set U = R.carrier;  
set e = algebra.group.identity(R qua Group);
```

Reason:

- Ambiguous punctuation must not make migration depend on parser guesses.

Example:

```
let N be Ring;  
let U = N carrier;  
let s = migration_group, side_group, group;
```

Reason:

- Ambiguous punctuation must not make migration depend on parser guesses.

Presenter script: Read aloud:

- Ambiguous punctuation must not make migration depend on parser guesses.

After the example, say which design obligation the syntax makes visible.

Grammar sketch:

```

type_expression ::= attribute_chain type_head ;
type_head       ::= radix_type | mode_type ;

attribute_chain ::= { [ "non" ] attribute_ref } ;
attribute_ref   ::= [ param_prefix ]
                   [ struct_ref_name "." ] attribute_name
                   [ "(" argument_list ")" ] ;

radix_type      ::= "object" | "set"
                  | struct_ref_name [ type_args ] ;

```

Grammar sketch:

```

mode_type       ::= mode_ref_name [ type_args ] ;

```



```

Grammar sketch:
| type_expression ::= attribute_chain type_bound ;
| type_bound      ::= mod_in_type | mode_type ;

attribute_chain ::= ( ( "and" | attribute_ref ) ;
attribute_ref  ::= ( param_ref in |
                    | struct_ref_name ".*" | attribute_name
                    | "[* argument_list]" ) ;

mod_in_type    ::= "object" | "set"
                | struct_ref_name | type_name ;

```

```

Grammar sketch:
mode_type      ::= mode_ref_name | type_name ;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Type Expression Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Example:

```
reserve G for non empty Group;  
let x be Element of G;  
let H be commutative subgroup of G;
```

Review Point:

- Mizar's soft type style is preserved, but the grammar makes the split between radix type, mode type, and attribute chain explicit.

Example:

```
preserve G for non empty Group;  
let x be Element of G;  
let H be commutative subgroup of G;
```

Review Point:

- Mizar's soft type style is preserved, but the grammar makes the split between radix type, mode type, and attribute chain explicit.

Presenter script: Read aloud:

- Mizar's soft type style is preserved, but the grammar makes the split between radix type, mode type, and attribute chain explicit.

After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```

definition_block    ::= "definition"
                        { definition_content }
                        "end" ";" ;

definition_content ::= { annotation }
                    ( definition_parameter_decl
                    | assumption
                    | correctness_condition
                    | property_item
                    | [ visibility ] definitional_item

```

Grammar sketch:

```

                    | [ visibility ] theorem_item
                    | [ visibility ] registration_item ) ;

```

```

Grammar <block>:
def in th in_n_block ::= "def init item"
    { def in th in_n_content }
    "end" ";" ;

def in th in_n_content ::= {
    nonterm in_n
    | def in th in_n_precondition, decl
    | nonterm in_n
    | correctness_condition
    | property_invariant
    | visibility | definition_item
}

```

```

Grammar <block>:
    | visibility | theorem_item
    | visibility | registration_item | ;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Definition Block Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Example:

```
definition
  let G be Group;
  func UnitDef: unit(G) -> Element of G
    means for x being Element of G holds it * x = x;
  existence by group.unit_exists;
  uniqueness by group.unit_unique;
end;
```

Review Point:

- The definition ... end envelope remains familiar.
- New surface elements should be judged by whether they clarify obligations without breaking Mizar's definition-oriented reading style.

Example:

```

def in G is in
  let G be Group;
  prove the G is unit(G) => Element of G
  mean for x being Element of G hold it = x * x;
  existence by group.unit_exists;
  uniqueness by group.unit_unique;
end;

```

Review Point:

- The def ID is ID ... end envelope remains familiar.
- New surface elements should be judged by whether they clarify obligations without breaking Mizar's definition-oriented reading style.

Presenter script: Read aloud:

- The definition ... end envelope remains familiar.
- New surface elements should be judged by whether they clarify obligations without breaking Mizar's definition-oriented reading style.

After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
struct_def      ::= "struct" struct_def_name [ type_params ]  
                  "where" struct_member { struct_member }  
                  "end" ";" ;  
  
struct_member   ::= field_decl | property_decl ;  
field_decl      ::= "field" identifier "->" type_expression  
                  [ "!=" term_expression ] ";" ;  
property_decl   ::= "property" identifier "->" type_expression ";" ;
```


└ Part 2. Language Specification

└ 2.13 - Structure Declaration Syntax (1/2)

Grammar sketch:

```
struct_decl ::= "struct" struct_def_name [ type_params ]  
            "where" struct_member ( struct_member )  
            "end" ";"  
  
struct_member ::= field_decl [ property_decl ]  
field_decl ::= "field" identifier ">" type_expression  
            [ ":" term_expression ] ";"  
property_decl ::= "property" identifier ">" type_expression ";"
```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Structure Declaration Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Example:

```
struct UnitalMagmaStr where
  field carrier -> non empty set;
  field mult -> BinOp of carrier;
  property unit -> Element of carrier;
end;
```

Review Point:

- field declares intrinsic data of the structure value.
- property declares a canonical derived slot or obligation.
- The syntax deliberately separates layout from mathematically determined values.

Example:

```

structure: This is an Mizar structure
field carrier -> an empty set;
field mult -> 0 (multiplication);
property zero -> element of carrier;
end;

```

Review Points:

- field declares intrinsic data of the structure value.
- property declares a canonical derived slot or obligation.
- The syntax deliberately separates layout from mathematically determined values.

Presenter script: Read aloud:

- field declares intrinsic data of the structure value.
- property declares a canonical derived slot or obligation.
- The syntax deliberately separates layout from mathematically determined values.

After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
property_means_impl ::= "property" identifier "." identifier  
                    "means" formula_definiens ";"  
                    existence_block uniqueness_block ;  
property_equals_impl ::= "property" identifier "." identifier  
                       "equals" term_definiens ";" ;
```

└ Part 2. Language Specification

└ 2.14 - Property Implementation Syntax (1/3)

Grammar sketch:

```
prophecy_exists_impl ::= "property" identifier "." identifier  
                        "exists" identifier_in_scope ":"  
                        existence_block uniqueness_block ;  
prophecy_equals_impl ::= "property" identifier "." identifier  
                        "equals" term_definition ":" ;
```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Property Implementation Syntax (1/3). Point to the visible example, question, or diagram before moving on.

Example:

```
definition
  let M be UnitalMagma;
  property M.unit means
    for x being Element of M.carrier holds
      M.mult(it, x) = x & M.mult(x, it) = x;
  existence by magma.unit_exists;
  uniqueness by magma.unit_unique;
end;
```

Example:

```
def is it in
  let M be Multiset;
  property M is it in
    for x being Element of M carrier hold
      M.mult(x) = x & M.mult(x, it) = x;
  existence by mset.is_it_in;
  uniqueness by mset.is_it_in_uniq;
end;
```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Property Implementation Syntax (2/3). Point to the visible example, question, or diagram before moving on.

Review Point:

- A property in a `struct` is a virtual slot, not stored field data.
- The implementation is supplied later for a mode whose attributes guarantee the required assumptions.
- `means` carries existence and uniqueness; `equals` directly supplies the value.

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- A property in a struct is a virtual slot, not stored field data.
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- A property in a struct is a virtual slot, not stored field data.
- The implementation is supplied later for a mode whose attributes guarantee the required assumptions.
- means carries existence and uniqueness; equals directly supplies the value.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
inherit_def ::= "inherit" inherit_child "extends" parent_type  
             [ "where" inherit_member { inherit_member }  
             [ coherence_block ] "end" ] ";" ;  
  
field_redef  ::= "field" identifier [ "->" type_expression ]  
             "from" ( identifier | "it" ) ";" ;  
property_redef ::= "property" identifier [ "->" type_expression ]  
             "from" identifier ";" ;
```

└ Part 2. Language Specification

└ 2.15 - Inheritance Declaration Syntax (1/2)

Grammar sketch:

```
inherit_def ::= "inherit" inherit_child "extends" parent_type
            | "extends" inherit_parent { inherit_child }
            | coherence_block [ "end" ] ";" ;

if_inherit_def ::= "if" id "inherit" id of id in { ">" type_expression |
            "true" | identifier } "is" ";" ;

pre_post_inherit_def ::= "pre" pre_id "inherit" id { ">" type_expression in }
            "is" id "inherit" ";" ;
```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Inheritance Declaration Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Example:

```
inherit GroupStr extends UnitalMagmaStr where
  field carrier from carrier;
  field mult from mult;
  property unit from unit;
  coherence by group.inherit_unital_magma;
end;
```

Review Point:

- Each `inherit` statement names exactly one parent.
- Renaming, narrowing, and coherence are visible source items.
- Diamond inheritance becomes a checked mapping problem rather than a hidden merge.

Example:

```
inherit M_Compiler extends M_Interpreter where  
  field carrier from carrier;  
  field unit from unit;  
  preserve unit from unit;  
  coherence by group, inherit_unit, merge;  
end;
```

Review Point:

- Each inherit statement names exactly one parent.
- Renaming, narrowing, and coherence are visible source items.
- Diamond inheritance becomes a checked mapping problem rather than a hidden merge.

Presenter script: Read aloud:

- Each inherit statement names exactly one parent.
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- Diamond inheritance becomes a checked mapping problem rather than a hidden merge.

After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
attr_def ::= "attr" label ":" subject "is" attr_pattern
           "means" formula_definiens ";" ;

mode_def ::= "mode" label ":" mode_def_name [ type_params ]
           "is" attribute_chain radix_type ";"
           [ mode_property ] ;
```

Example:

```
attr AssocDef:
  G is associative means
    for x, y, z being Element of G holds (x * y) * z = x * (y * z);

mode GroupDef: Group is associative invertible UnitalMagmaStr;
```

Grammar sketch:

```

attr_def ::= "attr" label ":" subject "is" attr_pattern
          "means" formal_def clause ";" ;

mode_def ::= "mode" label ":" mode_def_name : type_name {
          "is" attribute_chain mode_type ";"
          } mode_property ;

```

Example:

```

attr AssocDef:
  G is associative means
    for x, y, z being Element of G holds (x + y) + z = x + (y + z);

mode GroupDef: Group is associative invertible UnitRegister;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Attribute And Mode Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Review Point:

- Attributes are unary type-refining predicates.
- Modes name reusable type classifications.
- This keeps Mizar's adjective-rich prose while making cluster participation explicit.

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- Attributes are unary type-refining predicates.
- Modes name reusable type classifications.
- This keeps Mizar's adjective-rich prose while making cluster participation explicit.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
pred_def ::= "pred" label ":" pred_pattern
           "means" formula_definiens ";" ;

func_def ::= "func" label ":" func_pattern "->" type_expression
           ( "means" formula_definiens
             | "equals" term_definiens ) ";" ;
```

Example:

```
pred DividesDef:
  x divides y means ex z being Integer st y = x * z;

func InvDef: inverse(G, x) -> Element of G
  means x * it = unit(G) and it * x = unit(G);
```

└ Part 2. Language Specification

└ 2.17 - Predicate And Functor Syntax (1/2)

Grammar sketch:

```

pred_def ::= "pred" l_label ":" "pred_predicate"
           "meaning" formula_def in_line ";" ;

func_def ::= "func" l_label ":" "func_predicate" "<=>" type_expression
           | "name" formula_def in_line
           | "equals" term_def in_line | ";" ;

```

Example:

```

pred DividesDef :
  x divides y means x <= being Integer & y = x * z ;

func InvDef : inverse(G, x) => Element of G
  means x * it = unit(G) and it * x = unit(G) ;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Predicate And Functor Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Definition patterns:

- Predicate and functor patterns may use symbolic or phrase notation.
- Their symbols become active only after the declaration item is complete.
- Overloaded definitions are resolved by roots, argument types, and refinement rules, not by accidental source-order behavior.

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- Predicate and functor patterns may use symbolic or phrase notation.
- Their symbols become active only after the declaration item is complete.
- Overloaded definitions are resolved by roots, argument types, and refinement rules, not by accidental source-order behavior.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
synonym_def  ::= "synonym" alt_pattern "for" original_pattern ";" ;
antonym_def  ::= "antonym" alt_pattern "for" original_pattern ";" ;

redefine_attr ::= "redefine" "attr" label ":" subject "is" attr_pattern
                  "means" formula_definiens ";"
                  "coherence" [ "with" label ] justification ";" ;
redefine_pred ::= "redefine" "pred" label ":" pred_pattern
                  "means" formula_definiens ";"
                  "coherence" [ "with" label ] justification ";" ;
redefine_func ::= "redefine" "func" label ":" func_pattern
```

└ Part 2. Language Specification

└ 2.18 - Synonyms, Antonyms, And redefine (1/4)

Grammar sketch:

```

synonym_def ::= "synonym" alt_pattern "for" original_pattern ";" ;
antonym_def ::= "antonym" alt_pattern "for" original_pattern ";" ;

redefine_subj ::= "redefine" label ":" subject "is" alt_pattern
               "means" formal_definition ";" ;
"can be seen" : "is" label : justification ";" ;

redefine_pred ::= "redefine" label ":" pred_pattern
               "means" formal_definition ";" ;
"can be seen" : "is" label : justification ";" ;

redefine_func ::= "redefine" label ":" func_pattern

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Synonyms, Antonyms, And redefine (1/4). Point to the visible example, question, or diagram before moving on.

Grammar sketch:

```
"->" type_expression  
( "means" formula_definiens | "equals" term_definiens ) ";"  
"coherence" [ "with" label ] justification ";" ;
```


└ Part 2. Language Specification

└ 2.18 - Synonyms, Antonyms, And redefine (2/4)

Grammar sketch:

```
">" type_synonym  
| "same" formula_definition | "equals" term_definition | ":"  
"reference" | "rich" label | justification ":" ;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Synonyms, Antonyms, And redefine (2/4). Point to the visible example, question, or diagram before moving on.

Example:

```

definition
  let a, b be Real;
  pred LessDef: a < b means real.lt(a, b);
  synonym b > a for a < b;
  antonym a >= b for a < b;
end;

definition
  let x be non negative Real;
  redefine func AbsNonNeg: |.x.| -> non negative Real equals x;

```

Example:

```

  coherence with AbsGeneral by real.abs_nonneg;
end;

```

Example:

```
def in it is n
  let a, b be Real;
  pred lessDef: a < b means real.lt(a, b);
  synonym b > a for a < b;
  antonym a > b for a < b;
end;

def in it is n
  let x be non negative Real;
  redefine fun AbsNegDef: [x.] -> non negative Real equals x;
```

Example:

```
coherence with AbsGeneral by real.abs_negmg;
end;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Synonyms, Antonyms, And redefine (3/4). Point to the visible example, question, or diagram before moving on.

Review Point:

- Synonyms and antonyms preserve mathematical phrasing and natural negation.
- They are semantic aliases, not new roots.
- `redefine` sharpens the same root with `coherence with`; it is not a rival overload.

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Presenter script: Read aloud:

- Synonyms and antonyms preserve mathematical phrasing and natural negation.
- They are semantic aliases, not new roots.
- `redefine` sharpens the same root with coherence with; it is not a rival overload.

Grammar sketch:

```
pred_property ::= ( "symmetry" | "asymmetry" | "connectedness"  
                  | "reflexivity" | "irreflexivity" )  
                justification ";" ;  
  
func_property ::= ( "commutativity" | "idempotence"  
                  | "involutiveness" | "projectivity" )  
                justification ";" ;
```

Grammar sketch:

```
pred_property := | "symmetry" | "asymmetry" | "commutative"
               | "reflexivity" | "irreflexivity" |
               | "transitivity" ;
funct_property := | "associativity" | "idempotence"
                 | "invertibility" | "projectivity" |
                 | "isotonicity" ;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Predicate And Functor Properties (1/3). Point to the visible example, question, or diagram before moving on.

Example:

```
definition
  let x, y be Real;
  pred LessDef: x < y means x is_less_than y;
  asymmetry by real.lt_asym;
  irreflexivity by real.lt_irrefl;
end;
```

```
definition
  let X, Y be set;
  func UnionDef: X \ / Y -> set means
```


Example:

```
def is_it_in
  let x, y be Real;
  pred is_indef: x < y means x is_less_than y;
  asymmetry by real.lt_asym;
  reflexivity by real.lt_refl;
end;

def is_it_in
  let X, Y be set;
  func Unindef: X /> Y -> set means
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Predicate And Functor Properties (2/3). Point to the visible example, question, or diagram before moving on.

Example:

```
    for z being object holds z in it iff z in X or z in Y;  
    commutativity by set.union_comm;  
    idempotence by set.union_idem;  
end;
```

Review Point:

- Properties are not comments and not trusted hints.
- Each property is accepted only with a proof obligation.
- The design keeps algebraic notation compact while recording exactly which automatic matching, rewriting, and ATP facts may be used.

Example:

```

for z being object holds z in A iff z in X or z in Y;
connectivity by set, union, com;
subsequence by set, union, ides;
end;

```

Review Point:

- Properties are not comments and not trusted hints.
- Each property is accepted only with a proof obligation.
- The design keeps algebraic notation compact while recording exactly which automatic matching, rewriting, and ATP facts may be used.

Presenter script: Read aloud:

- Properties are not comments and not trusted hints.
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After the example, say which design obligation the syntax makes visible.

Key distinction:

Form	Meaning	Obligations
<code>equals</code>	gives a direct term definition	usually immediate well-definedness
<code>means</code>	characterizes a value or relation	may require existence, uniqueness, coherence, or compatibility
<code>assume</code>	adds a local definition-side condition	must be recorded as part of the generated obligation

└ Part 2. Language Specification

└ 2.20 - equals, means, And Correctness Blocks (1/3)

Key distinction:

Form	Meaning	Guidelines
equals	states a direct term definition	usually no modifier add state of term
means	characterizes a value or relation	may require occurrence, an operator, character, or computationality
correctness	states a local definition-side condition	may be recorded as part of the general definition

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: equals, means, And Correctness Blocks (1/3). Point to the visible example, question, or diagram before moving on.

Grammar sketch:

```
existence_block      ::= "existence" justification ";" ;  
uniqueness_block    ::= "uniqueness" justification ";" ;  
coherence_block     ::= "coherence" justification ";" ;  
compatibility_block ::= "compatibility" justification ";" ;
```

Example:

```
func InvDef: inverse(G, x) -> Element of G  
  means x * it = unit(G) and it * x = unit(G);  
existence by group.inverse_exists;  
uniqueness by group.inverse_unique;
```

└ Part 2. Language Specification

└ 2.20 - equals, means, And Correctness Blocks (2/3)

Grammar sketch:

```

existence_block  := "existence" just if used in " ; " ;
uniqueness_block := "uniqueness" just if used in " ; " ;
existence_block  := "existence" just if used in " ; " ;
compatibility_block := "compatibility" just if used in " ; " ;

```

Example:

```

func InvDef: inverse(G, x) => Element of G
means x = 1/x on G and 1/x = x on G;
existence by group.inverse_exists;
uniqueness by group.inverse_unique;

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: equals, means, And Correctness Blocks (2/3). Point to the visible example, question, or diagram before moving on.

Review Point:

- The syntax should preserve Mizar's definitional idiom while exposing the proof obligations that migration tools must not hide.

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Presenter script: Read aloud:

- The syntax should preserve Mizar's definitional idiom while exposing the proof obligations that migration tools must not hide.

Grammar sketch:

```
infix_operator_decl  ::= "infix_operator" "(" string_literal ","  
                        infix_assoc "," nat_literal ")" ";" ;  
prefix_operator_decl ::= "prefix_operator" "(" string_literal ","  
                        nat_literal ")" ";" ;  
postfix_operator_decl ::= "postfix_operator" "(" string_literal ","  
                        nat_literal ")" ";" ;  
infix_assoc          ::= "left" | "right" | "none" ;
```

└ Part 2. Language Specification

└ 2.21 - Operator Declarations And Precedence (1/4)

Grammar sketch:

```
inf ix_op precedence_decl ::= "inf ix_op precedence" "[" string_literal "]"  
                             "inf ix_op precedence" "[" string_literal "]"  
pre def op precedence_decl ::= "pre def op precedence" "[" string_literal "]"  
                             "pre def op precedence" "[" string_literal "]"  
post of op precedence_decl ::= "post of op precedence" "[" string_literal "]"  
                             "post of op precedence" "[" string_literal "]"  
inf ix_op precedence ::= "inf ix_op precedence" "[" string_literal "]"
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Operator Declarations And Precedence (1/4). Point to the visible example, question, or diagram before moving on.

Example:

```
infix_operator("+", left, 80);  
infix_operator("*", left, 90);  
infix_operator("^", right, 95);  
prefix_operator("-", 85);  
postfix_operator("!", 95);
```

```
a + b * c      :: a + (b * c)  
a ^ b ^ c      :: a ^ (b ^ c)  
a %% b %% c    :: error if %% is non-associative
```

Example:

```

inf ix_precedence "*" , left , 00 ;
inf ix_precedence "/" , left , 00 ;
inf ix_precedence "-" , right , 00 ;
pre def ix_precedence "+" , 00 ;
pre def ix_precedence "^" , 00 ;

a = b * c      :: a = b * c
a = b / c      :: a = b / c
a = b - c      :: a = b - c
a 00 to 00 c   :: error if 00 is non-associative

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Operator Declarations And Precedence (2/4). Point to the visible example, question, or diagram before moving on.

Design Reason:

- Term precedence and formula precedence are separate domains.
- Precedence values range from 0 to 255; higher values bind more tightly.
- A symbolic functor with no declaration defaults to precedence 64 and non-associative grouping.
- Operator declarations affect only later tokens; they are not forward declarations, and the target spelling must already be an active functor.
- Parsing chooses grouping from operator metadata; overload resolution later chooses the semantic root.

Design Reason:

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Presenter script: Read aloud:

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- Parsing chooses grouping from operator metadata; overload resolution later chooses the semantic root.

Design Reason:

- Conflicting imported metadata for the same visible user symbol is a link-time conflict.
- Built-in predicate symbols such as `=`, `<>`, and `in` come from the core module and cannot be overridden.
- `qua` is fixed by the language, has the lowest term precedence, and is left-associative.

Design Reason:

- Conflicting imported metadata for the same visible user symbol is a link-time conflict.
- Built-in predicate symbols such as $=$, $<$, $>$, and in come from the core module and cannot be overridden.
- qua is fixed by the language, has the lowest term precedence, and is left-associative.

Presenter script: Read aloud:

- Conflicting imported metadata for the same visible user symbol is a link-time conflict.
- Built-in predicate symbols such as $=$, $<$, $>$, and in come from the core module and cannot be overridden.
- qua is fixed by the language, has the lowest term precedence, and is left-associative.

Grammar sketch:

```

term_expression ::= operator_expression
                  { "qua" type_expression } ;

term_postfix     ::= "." field_name [ "(" [ term_list ] ")" ]
                  | "with" "(" field_update_list ")" ;

field_update     ::= selector ":=" term_expression ;

```

Example:

```

let x be object;
reconsider y = x qua Element of G;
set C = G.carrier;
set H = G with (carrier := C);

```

Grammar sketch:

```

term_expression ::= object_term
                  { "qua" type_expression } ;

term_prefix ::= "." d_term
             { "with" "}" d_term_update_list "}" ;

d_term_update ::= selector ":" term_expression ;

```

Example:

```

let x be object;
reconsider y = x qua Element of G;
let G = G.number;
let H = G with carrier := G;

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Term Expressions, Selectors, And qua (1/2). Point to the visible example, question, or diagram before moving on.

Review Point:

- Field/property access remains readable.
- qua remains source-level syntax because it records an intended view, not a failed inference.
- Selector syntax and namespace syntax must be resolved with source spans and explanation artifacts.

└ Part 2. Language Specification

└ 2.22 - Term Expressions, Selectors, And qua (2/2)

Review Point:

- Field/property access remains readable.
- qua remains source-level syntax because it records an intended view, not a failed inference.
- Selector syntax and namespace syntax must be resolved with source spans and explanation artifacts.

Presenter script: Read aloud:

- Field/property access remains readable.
- qua remains source-level syntax because it records an intended view, not a failed inference.
- Selector syntax and namespace syntax must be resolved with source spans and explanation artifacts.

Grammar sketch:

```
term_primary      ::= variable_identifier  
  
                  | "it"  
                  | numeral  
                  | "(" term_expression ")"  
                  | struct_constructor  
                  | set_expression  
                  | choice_expression  
                  | inline_functor_application
```

└ Part 2. Language Specification

└ 2.23 - Term Primary Forms: Constructors, Sets, And the (1/4)

Grammar sketch:

```

term_primary      := variable _ident id let
                    | "is"
                    | numeral
                    | "(" term_expression ")"
                    | set_expression
                    | set_application
                    | include_expression
                    | include_function_application

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Term Primary Forms: Constructors, Sets, And the (1/4). Point to the visible example, question, or diagram before moving on.

Grammar sketch:

```
term_primary      ::= variable_identifier  
  
                  | "it"  
                  | numeral  
                  | template_functor_application  
                  | bracket_functor_application ;
```


Grammar sketch:

```
term_primary      := var_table_id id id id  
                  | "is"  
                  | numeral  
                  | template_function_application  
                  | bracket1_function_application ;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Term Primary Forms: Constructors, Sets, And the (2/4). Point to the visible example, question, or diagram before moving on.

Grammar sketch:

```

struct_constructor ::= struct_name [ type_args ]
                    "(" [ field_assignment_list ] ")" ;
inline_functor_application ::= inline_func_name "(" [ term_list ] ")" ;
template_functor_application ::= functor_symbol template_args
                               [ "(" [ term_list ] ")" ] ;
bracket_functor_application ::= user_symbol term_list user_symbol
                              | "[" term_list "]" ;
set_expression      ::= "{" [ term_list ] "}"
                      | "{" term_expression "where"

```

Grammar sketch:

```

                    typed_var_list [ ":" formula ] "}" ;
choice_expression ::= "the" type_expression ;

```

Grammar sketch:

```

constructor ::= struct_name [ type_arg ]
              "(" [ field_name segment_list ] ")" ;
init_term_form_to_applicable_in ::= init_term_form_name "(" [ term_list ] ")" ;
template_term_form_to_applicable_in ::= constructor_name ( template_arg
              [ "(" term_list ] ")" ) ;
bracket_constructor_application ::= "(" var_symbol term_list var_symbol
              [ "(" term_list ] ")" ;
set_expression ::= "(" [ term_list ] ")"
              [ "(" term_expression "where"

```

Grammar sketch:

```

typed_var_list [ "(" domain ] ")" ;
init_term_expression ::= "the" type_expression ;

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Term Primary Forms: Constructors, Sets, And the (3/4). Point to the visible example, question, or diagram before moving on.

Example:

```
set p = Point(x: 3, y: 4);  
set evens = { n where n is Element of NAT : n mod 2 = 0 };  
set witness = the Element of X;
```

Review Point:

- These are ordinary mathematical terms, not algorithmic runtime constructs.
- the T depends on non-emptiness evidence for T.
- Set comprehensions are admitted only when the generator range is known to be a set.

Example:

```

set p = Pairs(x: 3, y: 4);
set even = { n where n is Even and NAT : n mod 2 = 0 };
set squares = {n | Element of X};

```

Review Point:

- These are ordinary mathematical terms, not algorithmic runtime constructs.
- that T depends on non-emptiness evidence for T .
- Set comprehensions are admitted only when the generator range is known to be a set.

Presenter script: Read aloud:

- These are ordinary mathematical terms, not algorithmic runtime constructs.
- the T depends on non-emptiness evidence for T .
- Set comprehensions are admitted only when the generator range is known to be a set.

After the example, say which design obligation the syntax makes visible.

Grammar sketch:

```
mode_property      ::= "sethood" justification ";" ;  
set_comprehension ::= "{" term_expression "where"  
                        typed_var_list [ ":" formula ] "}" ;
```

Example:

```
definition  
  mode FiniteOrdinalDef: FiniteOrdinal is ordinal finite set;  
  sethood by ordinal.finite_ordinal_sethood;  
end;  
  
set S = { f where f is FiniteOrdinal : f is non empty };
```

└ Part 2. Language Specification

└ 2.24 - sethood And Fraenkel Safety (1/2)

Grammar sketch:

```

node_property    ::= "sethood" justification "1";
set_comprehension ::= "{" term_1 opens in "where"
                      typed_var_1 int | ":" formula | "}" ;

```

Example:

```

def is finite in
  make FiniteOrdinDef: FiniteOrdinDef is ordinal finite set;
  sethood by ordinal.finite_ordinal_sethood;
end;

let S = { d where d is FiniteOrdinDef : d is non empty };

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: sethood And Fraenkel Safety (1/2). Point to the visible example, question, or diagram before moving on.

Review Point:

- sethood proves that all instances of a mode form a set, not a proper class.
- The type checker rejects $\{ f(x) \text{ where } x \text{ is } T : P(x) \}$ when no sethood witness or element-of-set range is available.
- This keeps the Tarski-Grothendieck set-theoretic boundary visible before ATP proof search begins.

Review Point:

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Presenter script: Read aloud:

- sethood proves that all instances of a mode form a set, not a proper class.
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- This keeps the Tarski-Grothendieck set-theoretic boundary visible before ATP proof search begins.

Grammar sketch:

```
formula          ::= quantified_formula | iff_formula ;
universal_formula ::= "for" quantified_vars [ "st" formula ]
                  ( "holds" formula | quantified_formula ) ;
existential_formula ::= "ex" quantified_vars "st" formula ;

iff_formula       ::= implies_formula
                  [ "iff" ( implies_formula | quantified_formula ) ] ;
implies_formula   ::= or_formula
                  [ "implies" ( implies_formula | quantified_formula ) ] ;
or_formula        ::= and_formula
```

Grammar sketch:

```

formula      ::= quantified_formula | iff_formula ;
universal_formula ::= 'forall' quantified_formula | 'ex' formula ;
exists_formula ::= 'exists' quantified_formula | 'not' formula ;
iff_formula   ::= implies_formula
               | 'iff' | implies_formula | quantified_formula | ;
implies_formula ::= formula
               | 'implies' | implies_formula | quantified_formula | ;
not_formula    ::= not_formula

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Formula Syntax (1/5). Point to the visible example, question, or diagram before moving on.

Grammar sketch:

```
                                { "or" ( and_formula | quantified_formula )
                                | "or" "... " "or"
                                  ( and_formula | quantified_formula ) } ;
and_formula ::= not_formula
              { "&" ( not_formula | quantified_formula )
              | "&" "... " "&"
                ( not_formula | quantified_formula ) } ;
not_formula  ::= "not" ( not_formula | quantified_formula )
```

Grammar sketch:

```
{ "x" | and_formula | quantified_formula |  
  "x" ... "x" }  
and_formula  := not_formula | quantified_formula |  
{ "x" | not_formula | quantified_formula |  
  "x" ... "x" }  
not_formula  := "not" | not_formula | quantified_formula |
```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Formula Syntax (2/5). Point to the visible example, question, or diagram before moving on.

Grammar sketch:

```

        { "or" ( and_formula | quantified_formula )
          | "or" "... " "or"
            ( and_formula | quantified_formula ) } ;
| atomic_formula
| "(" formula ")"

```

Grammar sketch:

```

        | "contradiction"
        | "thesis" ;

atomic_formula ::= predicate_application
                | inline_predicate_application
                | is_assertion ;

is_assertion  ::= term_expression "is" [ "not" ]

```

```
Grammar sketch:
{ "x" | not_formula | quantified_formula |
  | "x" "... " "x"
  | not_formula | quantified_formula | } ;
atomic_formula
  "[* formula "]"
```

```
Grammar sketch:
  "condition"
  "thesis" ;

atomic_formula  := predicate_application
                 | inline_predicate_application
                 | is_assertion ;
is_assertion   := term_expression "is" | "not" ;
```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Formula Syntax (3/5). Point to the visible example, question, or diagram before moving on.

Grammar sketch:

```
| "contradiction"  
| "thesis" ;  
    is_assertion_body ;
```

Example:

```
for x being Element of G st x is invertible holds  
  ex y being Element of G st x * y = unit(G);
```


Grammar sketch:

```
"conclusion"
| "thesis" :
  is_assertion_body ;
```

Example:

```
for x being Element of G at x is invertible holds
ex y being Element of G at x + y = unit[G];
```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Formula Syntax (4/5). Point to the visible example, question, or diagram before moving on.

Review Point:

- Mizar-style quantified prose is preserved.
- Attribute assertions and type assertions intentionally share surface syntax until resolution classifies them.
- Repeated connective forms such as `& ... &` and `or ... or` are explicit formula syntax, not parser accidents.

Review Points:

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Presenter script: Read aloud:

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- Attribute assertions and type assertions intentionally share surface syntax until resolution classifies them.
- Repeated connective forms such as $\& \dots \&$ and $\text{or} \dots \text{or}$ are explicit formula syntax, not parser accidents.

Slow down: this is language-specification review, not only implementation detail.

Precedence domains:

Domain	Source of precedence	Review point
Terms	user operator metadata plus fixed term syntax	parse before overload
Atomic formulas	predicates, equality, membership, is assertions	bridge from terms to formulas
Formulas	fixed hierarchy for not, &, or, implies, iff, quantifiers	no user-defined formula precedence

└ Part 2. Language Specification

└ 2.26 - Term/Formula Boundary And Formula Precedence (1/3)

Precedence domains:		
Domain	Design rationale	Design goal
Terms	avoid explicit or implicit precedence	parse without ambiguity
Atomic formulas	precedence, equality, membership, is a question	bridge from terms to formulas
Formulas	fixed hierarchy for not, &, or, implies, iff, quantifiers	no non-default formula precedence

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Term/Formula Boundary And Formula Precedence (1/3). Point to the visible example, question, or diagram before moving on.

Example:

```
a + b = c + d  
:: atomic formula over two parsed terms
```

```
not x > 0 & y > 0  
:: (not (x > 0)) & (y > 0)
```

```
a or b implies c  
:: (a or b) implies c
```

```
a iff b iff c
```

Example:

```
:: error: iff is non-associative
```

Example:

```
a = b * c + d
:: atomic formula over two parsed terms

not x > 0 & y > 0
:: inst |x > 0| & |y > 0|

a & b implies c
:: (a & b) implies c

a iff b iff c
```

Example:

```
:: atom or iff is non-associative
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Term/Formula Boundary And Formula Precedence (2/3). Point to the visible example, question, or diagram before moving on.

Review Point:

- Atomic formulas are the boundary between term parsing and formula parsing.
- Predicate-chain notation such as $a < b < c$ is resolved at this boundary, not as term-operator associativity.
- This makes the grouping explainable in diagnostics and stable for AI edits.

Review Points:

- Atomic formulas are the boundary between term parsing and formula parsing.
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Presenter script: Read aloud:

- Atomic formulas are the boundary between term parsing and formula parsing.
- Predicate-chain notation such as $a < b < c$ is resolved at this boundary, not as term-operator associativity.
- This makes the grouping explainable in diagnostics and stable for AI edits.

Grammar sketch:

```

theorem_item    ::= [ theorem_status ] theorem_role
                  label_identifier ":" formula
                  [ justification ] ";" ;

theorem_status  ::= "open" | "assumed" | "conditional" ;
theorem_role    ::= "theorem" | "lemma" ;

```

Example:

```

open theorem cancellation_candidate:
  for a, b, c being Element of G st a * b = a * c holds b = c;

assumed lemma choice_profile:
  for X being non empty set holds ex x being object st x in X;

```

Grammar sketch:

```

theorem_item  ::= { theorem_status { theorem_item
                    label_item id sep ":" domain
                    { justification } "*" ;
theorem_status ::= "open" | "assumed" | "conditional" ;
theorem_rule  ::= "theorem" | "lemma" ;

```

Example:

```

open theorem cancellation_cancellation:
  for a, b, c being Element of G at a * b = a * c holds b = c;

assumed lemma cancellation:
  for X being non empty set holds ex x being object at x in X;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Theorem Item And Status Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Review Point:

- `open`, `assumed`, and `conditional` are source-visible because unfinished or policy-controlled material must not look like ordinary verified theorem material.
- Publication profiles can restrict which statuses are allowed.

- open, assumed, and conditional are source-visible because unfinished or policy-controlled material must not look like ordinary verified theorem material.
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Presenter script: Read aloud:

- open, assumed, and conditional are source-visible because unfinished or policy-controlled material must not look like ordinary verified theorem material.
- Publication profiles can restrict which statuses are allowed.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
proof      ::= "proof" reasoning "end" ;
reasoning  ::= { annotated_statement } ;
statement  ::= [ "then" ] linkable_statement
              | standalone_statement ;

conclusion ::= ( "thus" | "hence" ) proposition
              justification ";" ;
```

Example:

```
proof
  let x be Element of G;
  assume A1: x = unit(G);
  thus thesis by A1, group.unit_left;
end;
```

└ Part 2. Language Specification

└ 2.28 - Proof Statement Syntax (1/2)

Grammar sketch:

```

proof      ::= "proof" term using "end" ;
proving    ::= < formula > _statement > ;
statement  ::= [ "% has" ] ! < formula > _statement
              | at end of _statement ;
condition  ::= [ "% has" ] "hence" | "because" | "because it is"
              just if it is "is" ;

```

Example:

```

proof
  let x be Element of G;
  assume A1: x = unit G;
  then thus is by A1, group.unit_left;
end;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Proof Statement Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Preserved proof vocabulary:

- let, assume, given, consider, take;
- thus, hence, thesis, by;
- now ... end, hereby ... end, per cases, case, suppose;
- deffunc, defpred, and reconsider.

Review Point:

- The readable Jaśkowski skeleton is not optional. The new artifacts support tools, but the source proof should still read as Mizar mathematics.

Preserved proof vocabulary:

- let, assume, given, consider, take;
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Review Point:

- The readable Jaśkowski skeleton is not optional! The new artifacts support tools, but the source proof should still read as Mizar mathematics.

Presenter script: Read aloud:

- let, assume, given, consider, take;
- thus, hence, thesis, by;
- now ... end, hereby ... end, per cases, case, suppose;
- deffunc, defpred, and reconsider.
- The readable Jaśkowski skeleton is not optional. The new artifacts support tools, but the source proof should still read as Mizar mathematics.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
iterative_equality ::= [ label_identifier ":" ]  
                      term_expression "=" term_expression  
                      simple_justification  
                      ".*" term_expression simple_justification  
                      { ".*" term_expression simple_justification } ";" ;  
  
statement ::= [ "then" ] linkable_statement  
            | standalone_statement ;
```

Grammar sketch:

```
iteration_qualify ::= { label_ident id sep ":" }
term_expression ::= term_expression
simple_justification ::=
  { "-" term_expression in simple_justification } "-" ;
statement ::= "then" | { label_id statement
  | statement_statement ;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Proof Statement Details (1/2). Point to the visible example, question, or diagram before moving on.

Example:

```
A1: f.(x + y) = f.x + f.y by Additive
      . = g.x + f.y by A2
      . = g.x + g.y by A3;
```

```
assume A2: f = g;
then f.x + f.y = g.x + f.y by FuncEq;
```

Review Point:

- `. =` is a justified equality chain, not assignment.
- `then` records dependence on the immediately preceding statement.
- Statement-level `such that` and `and` create labeled assumptions, while formula-level `st` and `&` remain inside one formula.

Example:

```

A1:  $f \cdot (x + y) = f \cdot x + f \cdot y$  by Additivity;
     $\bullet$   $g \cdot x = f \cdot y$  by A2;
     $\bullet$   $g \cdot x = g \cdot y$  by A3;

assume A2:  $f = g$ ;
then  $f \cdot x + f \cdot y = g \cdot x + f \cdot y$  by PropEq;

```

Review Point:

- $\bullet$ $\bullet$ is a justified equality chain, not assignment.
- $\bullet$ then records dependence on the immediately preceding statement.
- Statement-level such that and create labeled assumptions, while formula-level st and & remain inside one formula.

Presenter script: Read aloud:

- $\bullet$ $\bullet$ is a justified equality chain, not assignment.
- $\bullet$ then records dependence on the immediately preceding statement.
- Statement-level such that and create labeled assumptions, while formula-level st and & remain inside one formula.

After the example, say which design obligation the syntax makes visible.

Grammar sketch:

```

justification ::= simple_justification | proof | computation_proof ;
simple_justification ::= [ "by" references ] ;

reference ::= label_identifier [ template_args ]
           | qualified_reference [ template_args ]
           | grouped_reference
           | bulk_reference ;

grouped_reference ::= namespace_path ".{"
                    grouped_item { "," grouped_item } "}" ;

```

Grammar sketch:

```

bulk_reference    ::= namespace_path ".*" ;

```

```

Grammar <block>:
| justification ::= simple_justification | proof | computation_proof ;
| simple_justification ::= " " "by" reference ;

reference ::= label_justification | template_name [
    | qualified_reference | template_name [
        | grouped_reference
        | label_reference ;

grouped_reference ::= name_path "." "("
    | template_name "(" " " grouped_item ">" ;

```

```

Grammar <block>:
| label_reference ::= name_path "." ">" ;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Citation And Reference Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Example:

```
thus thesis by group.unit_left, group.{assoc, inverse_left};  
then thesis by algebra.group.*;
```

Review Point:

- Citation repair remains a small source edit.
- Bulk/grouped forms must be convenient without making dependencies invisible; artifacts record the actually used facts.

Example:

```
then then is by group.unleft, group.unleft, ungroup.unleft);  
then then is by algebra.group.*;
```

Review Point:

- Citation repair remains a small source edit.
- Bulk/grouped forms must be convenient without making dependencies invisible; artifacts record the actually used facts.

Presenter script: Read aloud:

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After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
registration_block ::= "registration"  
                    { registration_content }  
                    "end" ";" ;  
  
registration_item ::= existential_registration  
                  | conditional_registration  
                  | functorial_registration  
                  | reduction_registration ;  
  
conditional_registration ::= "cluster" label ":"
```

└ Part 2. Language Specification

└ 2.31 - Registrations, Clusters, And Reductions (1/3)

2.31 - Registrations, Clusters, And Reductions | 3/3

Grammar sketch:

```

registration_block ::= "registration"
                    { registration_content }
                    "end" ";" ;

registration_item ::= existential_registration
                  | conditional_registration
                  | function_registration
                  | reduction_registration ;

conditional_registration ::= "cluster" label ":"
  
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Registrations, Clusters, And Reductions (1/3). Point to the visible example, question, or diagram before moving on.

Grammar sketch:

```
antecedent_adjectives "->"  
consequent_adjectives  
"for" type_expression ";"  
"coherence" justification ";" ;
```

Example:

```
registration  
  cluster nonempty_group:  
    associative unital -> non empty for Magma;  
  coherence by group.nonempty_from_unit;  
end;
```

└ Part 2. Language Specification

└ 2.31 - Registrations, Clusters, And Reductions (2/3)

Grammar sketch:

```
non_empty_unit ::= "()"
cluster ::= "cluster" non_empty_unit
coherence ::= "coherence" justification ";" ;
```

Example:

```
registrations
  cluster nonempty_group;
  non_empty_unit ::= "()" non_empty_unit;
  coherence by group, non_empty_unit;
end;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Registrations, Clusters, And Reductions (2/3). Point to the visible example, question, or diagram before moving on.

Review Point:

- Registrations remain a first-class Mizar strength.
- The syntax adds labels and traceability so automatic propagation is reviewable and import-filtered.

Review Point:

- Registration remains a first-class Mizar strength.
- The syntax adds labels and traceability so automatic propagation is reviewable and import-filtered.

Presenter script: Read aloud:

- Registrations remain a first-class Mizar strength.
- The syntax adds labels and traceability so automatic propagation is reviewable and import-filtered.

Grammar sketch:

```
reduction_registration ::= "reduce" label ":"  
                        term_expression "to" term_expression ";"  
                        "reducibility" justification ";" ;
```

Example:

```
registration  
  let n be Nat;  
  reduce NatAddZero: n + 0 to n;  
  reducibility by nat.add_zero;  
end;
```


Grammar sketch:

```
reduction_registration ::= "reduce" label ";"  
                        "let" expression "is" term_expression ";"  
                        "reducibility" just if term_is ";" ;
```

Example:

```
registration  
  let n be Nat;  
  reduce NatAddTerm: n + 8 is n;  
  reducibility by nat_add_term;  
end;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Reduction Registration Semantics (1/2). Point to the visible example, question, or diagram before moving on.

Design Reason:

- A reduction is an oriented simplification rule backed by an equality proof.
- The right-hand side must be strictly smaller under the language's simplification order, so imported rules cannot create rewrite cycles.
- Normalization is deterministic: rule selection is specificity-first, with an FQN tie-break when needed.
- The classic unoriented `identify` idea is not used here; a simplifying equivalence must be written as an auditable `reduce`.

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- Normalization is deterministic: rule selection is specificity-first, with an FQN tie-break when needed.
- The classic unoriented identify idea is not used here; a simplifying equivalence must be written as an auditable reduce.

Grammar sketch:

```
template_definition ::= definition_block ;  
template_parameter_decl ::= definition_parameter_decl ;  
  
let_type ::= "type" [ "extends" bound_type ]  
           | type_expression ;  
  
template_item ::= attr_def | mode_def | struct_def  
               | pred_def | func_def | algorithm_def  
               | theorem_item | registration_item ;
```

Grammar sketch:

```
template_args ::= "[" template_arg { "," template_arg } "]" ;
```

Grammar sketch:

```

template_definition ::= def init_block ;
template_parameter_decl ::= def init_block_parameter_decl ;

let_type ::= "type" [ "extends" base_type ]
            type_expression ;

template_item ::= attr_def | mode_def | struct_def
              | proc_def | func_def | algorithm_def
              | theorem_item | registered_item ;

```

Grammar sketch:

```

template_arg ::= "!" template_arg ( "!" template_arg ) "*" ;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Template Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Example:

```
definition
  let T be type;
  struct MagmaStr[T] where
    field carrier -> T;
    field binop -> BinOp of T;
  end;
end;
```

Review Point:

- Templates are not a cosmetic replacement for scheme.
- They are the common syntax for parameterized definitions, theorem schemas, structures, registrations, and algorithms.
- of and over remain readable shorthands; bracket form is canonical for identity and tooling.

Example:

```
def isPrime
  let T be type;
  assume: Nondet(T); where
    f is id carrier -> T;
  end;
end;
```

Review Point:

- Templates are not a cosmetic replacement for schemes.
- They are the common syntax for parameterized definitions, theorem schemas, structures, registrations, and algorithms.
- `if` and `over` remain readable shorthands; bracket form is canonical for identity and tooling.

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- Templates are not a cosmetic replacement for scheme.
- They are the common syntax for parameterized definitions, theorem schemas, structures, registrations, and algorithms.
- `if` and `over` remain readable shorthands; bracket form is canonical for identity and tooling.

After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Grammar sketch:

```
template_args ::= "[" template_arg { "," template_arg } "]" ;  
let_constraint ::= "such" "that" formula ;  
let_type      ::= "type" [ "extends" bound_type ]  
                | type_expression ;
```


└ Part 2. Language Specification

└ 2.34 - Template Constraints And Inference (1/3)

Grammar sketch:

```
template_arg ::= "(" template_arg ( "," template_arg ) "*" ;  
val_constraint ::= "var" "has" formula ;  
val_type ::= "type" ( "outside" bound_type |  
                | type_expression ) ;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Template Constraints And Inference (1/3). Point to the visible example, question, or diagram before moving on.

Example:

```

definition
  let F be Field;
  let V be VectorSpace[F] such that V is finite_dimensional;
  mode BasisDef: Basis[V] is finite linearly_independent Subset of V.carrier;
end;

Product[Ring](s)      :: explicit template argument
Product(s)            :: valid only when the argument type determines Ring
M of A, B             :: greedy shorthand for M[A, B] when arity matches

```

Example:

```

definition
  let F be Field;
  let F be VectorSpace/F such that F is finite_dimensional;
  make assumed: usual/F is finite_linearly_independent Subset of F.vectors;
end;

Product1 is legal
:: explicit template argument
Product1
:: valid only when the argument type determines Ring
F of n, R
:: greedy shorthand for R is, R when using macros

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Template Constraints And Inference (2/3). Point to the visible example, question, or diagram before moving on.

Review Point:

- Bracket arguments are canonical because they preserve identity for tools.
- `of` and `over` are kept as readable shorthands, but greedy parsing must be deterministic.
- A `such that` constraint is a use-site obligation, not a hidden global assumption.

Review Point:

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Presenter script: Read aloud:

- Bracket arguments are canonical because they preserve identity for tools.
- of and over are kept as readable shorthands, but greedy parsing must be deterministic.
- A such that constraint is a use-site obligation, not a hidden global assumption.

Example:

```
@show_type(total + x)

@show_resolution
Product[R](s);

@proof_hint(max_axioms: 32, solver: vampire)
thus result = unit(G) by group.identity_unique;

@latex("\\mathbb{Z}")
func IntSetDef: INT -> set equals IntegerSet;
```

Part 2. Language Specification

└ 2.35 - Annotation Syntax (1/2)

Example:

```

@show_type{total * x}

@show_result{in
Product{in{1}};

@sprod_hint{max_exhausts: 33, solver: vampire}
this result = unit{G} by group.ident{by, unique};

@date{x} "%a %b %d %Y"}
@date OutputDir: OUT -> set equals IntegerSet;

```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Annotation Syntax (1/2). Point to the visible example, question, or diagram before moving on.

Review Point:

- Annotations are source syntax, but they must be semantic-neutral unless a specific proof-development profile treats them as hints for search.
- They guide editors, documentation, proof search, and AI agents.
- They do not become proof evidence merely by being present in the source.
- Fixed diagnostic annotations such as `@show_type(...)` and `@show_resolution` report state; they do not attach logical content to a declaration.

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- Fixed diagnostic annotations such as `@show_type(...)` and `@show_resolution_report state`; they do not attach logical content to a declaration.

Slow down: this is language-specification review, not only implementation detail.

Example:

```
definition
  let x, y be Integer;

  algorithm max2(x, y) -> Integer
    ensures (result = x or result = y) & x <= result & y <= result
  do
    if x <= y do
      return y;
    end;
    return x;
```

Example:

```
    end;
  end;
```

Example:

```
def isEven
  let x, y be Integer;

  algorithm even2(x, y) → Integer
  ensures |result = x or result = y| & x <= result & y <= result
  do
    if x <= y then
      return y;
    end;
    return x;
```

Example:

```
end;
end;
```

Presenter script: After the example, say which design obligation the syntax makes visible.

Slow down: this is language-specification review, not only implementation detail.

Use this as the transition: Algorithm Syntax Boundary (1/2). Point to the visible example, question, or diagram before moving on.

Review Point:

- Algorithms are part of the language surface, but they are not allowed to redefine theorem truth.
- They are declared inside `definition` blocks, and the public interface is the signature plus contracts, not the body.
- Contracts, invariants, termination measures, and `by computation` generate verification obligations that pass through the same trusted boundary.

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Slow down: this is language-specification review, not only implementation detail.

Questions for Bialystok:

- Which legacy Mizar constructs must keep near-identical syntax for cultural and migration reasons?
- Which constructs may become more explicit because the current implicit form hides dependencies?
- Are `field` / `property` / `attribute` distinctions readable enough in real algebraic examples?
- Is `one-parent-per-inherit` acceptable for multiple-inheritance-heavy MML fragments?
- Do the explicit operator-precedence rules match the expectations of existing Mizar notation users?

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Slow down: this is language-specification review, not only implementation detail.

Questions for Bialystok:

- Are `sethood` checks visible enough for Fraenkel-style set formation without making ordinary set notation too heavy?
- Which predicate/functor properties should be migrated first because existing proofs rely on them implicitly?
- Is oriented `reduce` a readable replacement for implicit simplification and older identification idioms?
- Are template brackets acceptable as canonical identity syntax if `of` and `over` remain source shorthands?
- Where should template inference stop and require explicit `[T]` arguments or `qua` views?

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- Which proof statement forms are missing from this surface review?

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- Where should template inference stop and require explicit [T] arguments or qua views?

Slow down: this is language-specification review, not only implementation detail.

- Current: `environ` gives each article its vocabulary, notation, constructors, registrations, requirements, and visible theorem base.
- Evo: the same roles must become explicit imports, prelude policy, package metadata, and reproducible dependency reports.
- Review question: which current `environ` roles must remain visually familiar during migration, and which can be replaced by generated reports?

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- Evo: export, public, private, opaque imports, and separate compilation define the reusable API surface.
- Review question: which legacy article facts should become public module API, and which should become private proof support?

└ Part 2. Language Specification

└ 2.39 - Detail: Module Interfaces

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- Current: theorem and constructor identity is tied to article names, labels, and MML inclusion history.
- Evo: path-derived FQNs and module-qualified labels make identity independent of the local import spelling.
- Review question: what identity metadata is needed to preserve old citations while allowing package/module reorganization?

└ Part 2. Language Specification

└ 2.40 - Detail: Symbol Identity

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- Review question: what identity metadata is needed to preserve old citations while allowing package/module reorganization?

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- Current: Mizar supports rich mathematical notation, including synonyms and antonyms for alternative phrasing and natural negation.
- Evo: arbitrary operator symbols are concentrated in `func` and `pred`, while `mode`, `attribute`, and structure names remain identifier-like; synonym and antonym equivalence is preserved.
- Review question: which legacy symbolic constructors need compatibility spelling, and which should migrate to clearer constructor names?

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Presenter script: Read aloud:

- Current: Mizar supports rich mathematical notation, including synonyms and antonyms for alternative phrasing and natural negation.
- Evo: arbitrary operator symbols are concentrated in func and pred, while mode, attribute, and structure names remain identifier-like; synonym and antonym equivalence is preserved.
- Review question: which legacy symbolic constructors need compatibility spelling, and which should migrate to clearer constructor names?

- Current: an article environment determines which symbols and notations are available, with many effects hidden from local text.
- Evo: the import prelude is pre-scanned, the active lexicon is source-position dependent, and declarations become active only after their item is complete.
- Review question: where will migration need generated diagnostics for no-forward-reference or dot/operator disambiguation changes?

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- Current: Mizar's soft type system is a central identity feature, layered over set-theoretic objects.
- Evo: soft typing is preserved while `object/set`, `radix` and `mode heads`, `type erasure`, `widening/narrowing`, and `reconsider obligations` are exposed.
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- Evo: `struct`, `field`, `property`, and `inherit` separate layout, canonical derived data, and inherited views.
- Review question: when migrating a legacy selector, is it intrinsic structure data, a derived property, or an inherited view?

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- Evo: each `inherit` statement records one parent relation, including mapping, renaming, narrowing, and coherence evidence.
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- Current: Jaśkowski-style proof text with `let`, `assume`, `thus`, `hence`, `thesis`, and diffuse reasoning is central to Mizar readability.
- Evo: the readable skeleton is preserved while extracted obligations, thesis states, and source spans are emitted as metadata.
- Review question: how much proof-state metadata helps users without making the proof look tactic-driven?

Mizar Evo

└ Part 2. Language Specification

└ 2.51 - Detail: Declarative Proof Skeleton

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- Evo: publication metadata links articles to reusable library modules, package versions, and stable theorem identities.
- Review question: how can Evo preserve citation value while letting the reusable library evolve independently?

└ Part 2. Language Specification

└ 2.63 - Detail: Formalized Mathematics Link

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Exact current MML excerpt:

```
environ
```

```
vocabularies XBOOLE_0, SUBSET_1, BINOP_1, ZFMISC_1, STRUCT_0, ARYTM_3,  
             FUNCT_1, FUNCT_5, SUPINF_2, ARYTM_1, RELAT_1, MESFUNC1, ALGSTR_0, CARD_1;  
notations TARSKI, XBOOLE_0, SUBSET_1, ZFMISC_1, BINOP_1, FUNCT_5, ORDINAL1,  
          CARD_1, STRUCT_0;  
constructors BINOP_1, STRUCT_0, ZFMISC_1, FUNCT_5;  
registrations ZFMISC_1, CARD_1, STRUCT_0;  
theorems STRUCT_0;
```

Exact current MML excerpt:

```
begin :: Additive structures
```

Exact current MML excerpt:

[illegible]

Exact current MML excerpt:

begin :: Additive structure

Presenter script: Say:

- Source: current MML algstr_0.miz, lines 15-25.
- The point is not that `environ` is bad. The point is that dependency classes are article-level and need translation into explicit module/package data.

```
import std.algebra.structure.sorted;  
import std.function.binop;  
import std.algebra.structure.magma;
```

Main differences:

- file-scoped import prelude;
- package/module namespace;
- import closure as a build-system input;
- stable FQNs derived from path and module.

Mizar Evo

└ Part 2. Language Specification

└ 2.65 - Mizar Evo Import Prelude (1/2)

```
import at d, at go bna, at must use, a, sort d;  
import at d, function, to dump;  
import at d, at go bna, at must use, a, name;
```

Main differences:

- file-scoped import prelude;
- package/module namespace;
- import closure as a build-system input;
- static FQN is derived from path and module.

Presenter script: Read aloud:

- file-scoped import prelude;
- package/module namespace;
- import closure as a build-system input;

After the example, say which design obligation the syntax makes visible.

Design Reason:

- The prelude is deliberately file-scoped so lexing, disambiguation, and AI context extraction can build one stable imported base environment before reading the body.
- This makes dependency reports reproducible: a tool can say which import provided each symbol, notation, registration, and theorem.

Design Reason:

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2.66 - Migration Map: Environment To Imports (1/2)

Current Mizar role	Evo target
vocabularies	exported symbols and lexical metadata
notations	imported notation metadata
constructors	visible definitions and constructors
registrations	import-scoped registration index
requirements	package or prelude policy

└ Part 2. Language Specification

└ 2.66 - Migration Map: Environment To Imports (1/2)

Source: Mizar file	Env target
constants	reported symbols and local constants
parameters	reported symbols and constants
constructions	rule & definitions and constructions
definitions	reported symbols and constants
requirements	package or package parts

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Migration Map: Environment To Imports (1/2). Point to the visible example, question, or diagram before moving on.

2.66 - Migration Map: Environment To Imports (2/2)

Current Mizar role	Evo target
article labels	module-qualified theorem identities

└ Part 2. Language Specification

└ 2.66 - Migration Map: Environment To Imports (2/2)

Can be: <code>Import</code>	Env type:
with <code>Import</code>	with <code>Import</code> from <code>Imports</code>

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Migration Map: Environment To Imports (2/2). Point to the visible example, question, or diagram before moving on.

Key Points:

- Current environments mix several dependency roles.
- Some dependencies are semantic, some syntactic, some automation-facing.
- Migration needs reports explaining what each imported module contributes.
- The purpose of the new boundary is not cosmetic modernization. It is to make old implicit roles reviewable before they become package and artifact dependencies.

Key Points:

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Current plain-text ALGSTR_0 exposes the shape:

```
definition
  struct (1-sorted) addMagma (# carrier -> set, addF -> BinOp of the carrier
    #);
end;
```

```
definition
  exact {Succed: addRagm (R carrier -> set, add -> kindy of the carrier
  R);
end;
```

Presenter script: Say:

- Source: current MML algstr_0.miz, lines 37-40.
- This is a short exact excerpt; the Beamer slide can use the same snippet with attribution in speaker notes.

Read aloud:

- Current plain-text ALGSTR_0 exposes the shape:

```
definition
  struct AddMagma where
    field carrier -> set;
    field add -> BinOp of carrier;
  end;

  struct AddLoopStr where
    field carrier -> non empty set;
    field add -> BinOp of carrier;
    property zero -> Element of carrier;

  end;

  inherit AddMagma extends 1-sorted;
end;
```

Mizar Evo

└ Part 2. Language Specification

└ 2.69 - Evo Structure Example (1/2)

```

def in it is n
  struct AddMagm where
    field carrier -> set;
    field add -> BinOp of carrier;
  end;

  struct AddMagmExt where
    field carrier -> no empty set;
    field add -> BinOp of carrier;
    property term -> Element of carrier;
  end;

  inher it. AddMagm extends 1-sorted;
end;

```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Evo Structure Example (1/2). Point to the visible example, question, or diagram before moving on.

Claim:

- Evo makes field declarations and inheritance obligations explicit, while keeping the concept close to Mizar structures.
- Evo also separates `field` from `property`: fields are independent structure components, while properties are uniquely determined mathematical data or obligations associated with the structure.

Design Reason:

- Structure inheritance remains useful only if users can see which fields are shared, renamed, or required by a parent structure.
- Making those obligations explicit turns a migration risk into something the verifier, LSP, and reviewers can discuss.
- Separating fields from properties prevents constructor layout, selector identity, and mathematical laws from being treated as the same kind of dependency.

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Design distinction:

Concept	What it means	Why it is separate
field	intrinsic component supplied by the structure value	affects constructor shape, selector identity, extensional equality, and storage-like artifacts
property	uniquely determined value or obligation attached to the structure	affects semantic obligations, coherence, inheritance constraints, and proof/artifact fingerprints
attribute	predicate-style refinement such as <code>empty</code> , <code>trivial</code> , or <code>add-cancelable</code>	participates in registration and cluster propagation rather than structure layout

Part 2. Language Specification

2.70 - Why Separate Field And Property? (1/2)

Design distinction:

Design distinction:		
Concept / role	What it means In other components supplied by the microcode value	What it compares effects construction the value of function, an emotional quality, and storage for an effect
properties	notified detached in value of interface attached to the microcode	affects semantic interpretation, enhancement, inheritance combinations, and protocol/abstract interpretation
use/case	breakdown into information: such as temp, c, status, or tag+content+value	as the source of regularities and demonstrate propagation rather than microcode performance

Presenter script: Say:

- Source design vocabulary: `doc/spec/en/05.structures.md` distinguishes `field` from `property`; `doc/spec/en/06.attributes.md` defines attributes as type-refining predicates used by clustering.

For the table, read the row labels first, then the design reason.

Message:

- A carrier or operation is a field because changing it changes the object.
- A zero, unit, degree, or similar canonical value may be a property when it is determined by the mathematical structure and needs coherence evidence.
- An attribute is neither stored data nor a selector; it is a reusable logical refinement that automation can propagate.
- This split gives migration tools a better question to ask: "is this legacy item part of the structure's data, a canonical value with proof obligations, or a propagated predicate?"

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Presenter script: Read aloud:

- A carrier or operation is a field because changing it changes the object.
- A zero, unit, degree, or similar canonical value may be a property when it is determined by the mathematical structure and needs coherence evidence.
- An attribute is neither stored data nor a selector; it is a reusable logical refinement that automation can propagate.
- This split gives migration tools a better question to ask: "is this legacy item part of the structure's data, a canonical value with proof obligations, or a propagated predicate?"

Design distinction:

Declaration	What it owns	Why it is separate
<code>struct</code>	the local type constructor, fields, and properties	defines the object's own layout and selectors
<code>inherit</code>	the relation to one parent structure	records field/property mapping, renaming, narrowing, and coherence evidence

└ Part 2. Language Specification

└ 2.71 - Why Separate struct And inherit? (1/2)

Design distinction:		
Declaration	Why it runs	Why it is separate
struct	the local type constructs, fields, and properties	the new the object's own type and methods
inherit	the relation to one parent or more	records full property mapping, naming, narrowing, and coherence evidence

Presenter script: Say:

- Source design vocabulary: `doc/spec/en/05.structures.md` requires one parent per inherit statement and uses explicit where blocks for renaming, narrowing, and coherence.

For the table, read the row labels first, then the design reason.

Message:

- A structure declaration should answer "what is this object made of?"
- An inheritance declaration should answer "how is this object viewed as that parent?"
- Keeping them separate makes multiple inheritance explicit: each parent gets one auditable `inherit` statement instead of one large implicit merge.
- Diamond inheritance then becomes a diagnostic problem with source spans and coherence obligations, not a hidden side effect of declaration order.
- It also gives artifacts cleaner fingerprints: changing a field layout is not the same event as changing an inherited view or its coherence proof.

└ Part 2. Language Specification

└ 2.71 - Why Separate struct And inherit? (2/2)

Message:

- A structure declaration should answer: "what is this object made of?"
- An inheritance declaration should answer: "how is this object viewed as that parent?"
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- Diamond inheritance then becomes a diagnostic problem with source spans and coherence obligations, not a hidden side effect of declaration order.
- It also gives artifacts cleaner fingerprints: changing a field layout is not the same event as changing an inherited view or its coherence proof.

```
definition
  struct Magma where
    field carrier -> set;
    field binop -> BinOp of carrier;
  end;

  inherit AddMagma extends Magma where
    field carrier from carrier;
    field add from binop;
  end;

end;
```



```
def in M is
  struct Magma where
    field carrier -> set;
    field binop -> BinOp of carrier;
  end;

  lemma M: AddMagma extends Magma where
    field carrier from carrier;
    field add from binop;
  end;
end;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Renaming And Views (1/2). Point to the visible example, question, or diagram before moving on.

Purpose:

- Support library organization where additive and multiplicative views share structure without relying on hidden informal naming conventions.
- Preserve the mathematical habit of reusing the same carrier through different views, while recording the view translation as checked source.

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- Support library organization where additive and multiplicative views share structure without relying on hidden informal naming conventions.
- Preserve the mathematical habit of reusing the same carrier through different views, while recording the view translation as checked source.

```
inherit DoubleLoopStr extends AddLoopStr;  
inherit DoubleLoopStr extends MulLoopStr;
```

Message:

- The system should explain when two inheritance paths agree, conflict, or need a proof of coherence.
- This explanation should be available to users, LSP, and AI agents.
- The design goal is to keep multiple-view algebra usable without letting hidden inheritance order decide semantics.

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After the example, say which design obligation the syntax makes visible.

Exact current MML excerpt:

```
registration
  let M be addMagma;
  cluster right_add-cancelable left_add-cancelable -> add-cancelable for
Element
  of M;
  coherence;
end;
```

Evo interpretation:

- registration is still a mathematical mechanism;
- the implementation stores an import-filtered, explainable registration graph;
- every automatic attribute propagation should be traceable.

Exact current MML excerpt:

```

registration
  let N be add@magm;
  cluster = right_half-ance table left_half-ance table -> add-ance table for
  N in magm;
  coherence;
end;

```

Evo interpretation:

- registration is still a mathematical mechanism;
- the implementation stores an import-filtered, explainable registration graph;
- every automatic attribute propagation should be traceable.

Presenter script: Say:

- Source: current MML `algstr_0.miz`, lines 104-109.
- The excerpt is intentionally short: it shows attribute propagation without making the slide depend on a long registration proof.

Read aloud:

- registration is still a mathematical mechanism;
- the implementation stores an import-filtered, explainable registration graph;
- every automatic attribute propagation should be traceable.

Design Reason:

- Registrations and clusters are one of Mizar's strengths, so Evo should not remove them.
- The change is that automatic propagation must leave a trace that can be audited, cached, and shown to an AI agent without dumping the whole library.

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```
consider p be Product(R qua MulStr) such that  
...
```

Message:

- Explicit disambiguation is not a failure of inference.
- It is a readable record of the intended mathematical view.
- It gives AI agents a small safe repair target.

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consider p be Product (R qua Mult1r) such that  
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Presenter script: Read aloud:

- Explicit disambiguation is not a failure of inference.
- It is a readable record of the intended mathematical view.
- It gives AI agents a small safe repair target.

After the example, say which design obligation the syntax makes visible.

Before:

```
thus thesis by A;
```

After:

```
thus thesis by A, B, C;
```

Before:

After:

Presenter script: Say:

- Source: current MML struct_0.miz, lines 637-643.
- Use this to explain citation repair as a bounded local edit: an agent may propose a missing or more precise citation, but the verifier and kernel evidence decide acceptance.

Exact current MML proof-citation example:

```
theorem
  for F being non degenerated ZeroOneStr holds 1.F in NonZero F
proof
  let F be non degenerated ZeroOneStr;
  not 1.F in {0.F} by TARSKI:def 1;
  hence thesis by XBOOLE_0:def 5;
end;
```

Message:

- This is a Green AI edit: source-local, verifier-checked, meaning-preserving if accepted with respect to the theorem statement.
- A later refinement pass can minimize citations based on used axioms recorded in artifacts.
- The added dependency still matters; artifacts should record it so a later mizar refine-style pass can reduce noise.

Exact current MML proof-citation example:

```

theorem
  for P being non degenerated Denotable holds 1.P in Members P
proof
  let P be non degenerated Denotable;
  not 1.P in (1.P) by TANUKI:def 1;
  hence thesis by XBOOLE_1:def 3;
end;

```

Message:

- This is a Green AI edit: source-local, verifier-checked, meaning-preserving if accepted with respect to the theorem statement.
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After the example, say which design obligation the syntax makes visible.

Bad AI repair:

```
theorem  
  for x be Nat holds  $x + 0 = x$  or  $x = 0$ ;
```

Message:

- Weakening the theorem statement is a Red edit.
- It may make the proof easier but destroys the intended result.
- Ordinary AI agents must not be authorized to apply it; at most they may flag that such a change would require explicit human unsafe-edit review.

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After the example, say which design obligation the syntax makes visible.

```
@show_type(total + x)

@show_resolution
Product[R](s);

@proof_hint(max_axioms: 32, solver: vampire)
thus result = unit(G) by group.identity_unique;
```

Message:

- Annotations are structured context for humans, tools, and AI.
- Informational annotations should not change logical meaning.
- Hints that influence search must still be treated as proof-development metadata, not as accepted evidence.

Part 2. Language Specification

└ 2.78 - Annotations

```
2.78 - Annotations

@show_type(typeof(x))

@show_result{m}
product{H1};

@prod{bind{max_extensions: 32, solver: vcsolve}}
class result = emit{G} by group_id{H1_y,unique};
```

Presenter script: Read aloud:

- Annotations are structured context for humans, tools, and AI.
- Informational annotations should not change logical meaning.
- Hints that influence search must still be treated as proof-development metadata, not as accepted evidence.

After the example, say which design obligation the syntax makes visible.

Proposed Evo sketch:

```
definition
  let T be type;
  struct MagmaStr[T] where
    field carrier -> T;
    field binop -> BinOp of T;
  end;
end;

definition
  let M be type extends commutative Magma;
```

```
def in it in
  let T be type;
  axiom: MapReduce(T) where
    f field carrier -> T;
  end;
end;

def in it in
  let M be type extends commutative Monoid;
```

Presenter script: Say:

- Source design vocabulary: `doc/spec/en/18.templates.md` defines templates as generics for parameterized definitions and theorem schemas, including structures, attributes, modes, predicates, functors, and schemes.

Proposed Evo sketch:

```
theorem PermProduct[M]:  
  ...  
end;
```

Message:

- Templates are the generics mechanism for Mizar Evo.
- They parameterize definitions and theorem schemas over types, predicates, or functors.
- Classical scheme-style reasoning becomes part of one template vocabulary rather than a separate special case.
- `of` / `over` forms remain readable shorthands; the bracket form records the canonical parameterization.

Proposed Evo sketch:

```

theorem Parameterized IM1 :
  ...
end

```

Message:

- Templates are the generics mechanism for Mizar Evo.
- They parameterize definitions and theorem schemas over types, predicates, or functors.
- Classical scheme-style reasoning becomes part of one template vocabulary rather than a separate special case.
- `def / over` form usually reads like shortcuts; the bucket form records the canonical parameterization.

Presenter script: Read aloud:

- Templates are the generics mechanism for Mizar Evo.
- They parameterize definitions and theorem schemas over types, predicates, or functors.
- Classical scheme-style reasoning becomes part of one template vocabulary rather than a separate special case.

After the example, say which design obligation the syntax makes visible.

Design Reason:

- Templates avoid copying the same algebraic construction for every carrier, field, or predicate instance.
- They make generic mathematical patterns explicit enough for dependency fingerprints, theorem indexes, and AI retrieval.
- Constraints such as `type extends commutative Magma` say what the parameter must provide before ATP or proof search begins.

Design Reason:

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Potential Evo sketch:

```
definition
  let x, y be Integer;

  algorithm max2(x, y) -> Integer
    ensures (result = x or result = y) & x <= result & y <= result
  do
    if x <= y do
      return y;
    end;
    return x;
```

Potential Evo sketch:

```
    end;
  end;
```

Potential Evo search:

```
def in it in
  let x, y be Integer;

  algorithm next(x, y) → Integer
  ensures {result = x or result = y} & x <= result & y <= result
  do
    if x <= y then
      return y;
    end;
    return x;
```

Potential Evo search:

```
end;
end;
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Algorithm Verification (1/2). Point to the visible example, question, or diagram before moving on.

Message:

- Algorithms broaden the scope beyond pure mathematics.
- Contracts and generated VCs must still feed the same verification pipeline.

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- Algorithms broaden the scope beyond pure mathematics.
- Contracts and generated VCs must still feed the same verification pipeline.

```
@[origin("mizar:ALGSTR_0:addMagma")]  
definition  
  struct AddMagma where  
    ...  
  end;  
end;
```

Message:

- Migration must preserve historical identity.
- Origin metadata supports Formalized Mathematics links, citation stability, and review.

```

@ is origin "mizar:ALGEBRA:addition" ()
def in the
  a term: Addition where
    ...
  end;
end;
end;

```

Message:

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Presenter script: Read aloud:

- Migration must preserve historical identity.
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After the example, say which design obligation the syntax makes visible.

Questions:

- Which legacy constructs require direct syntax preservation?
- Which constructs should be translated into clearer Evo forms?
- What origin identifiers are stable enough for MML migration?
- Which current environment dependencies are most difficult to explain?

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Source

- > TokenStream
- > SurfaceAst
- > ResolvedAst
- > TypedAst
- > CoreIr
- > VcIr
- > AtpProblem
- > ProofCertificate
- > VerifiedArtifact

```
Source
-> ToBackend
-> SurfacedAs
-> ResolvedAs
-> TypedAs
-> Lower
-> Verify
-> Allocate
-> FinalizeAs
-> VerifierAs
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Pipeline Overview (1/2). Point to the visible example, question, or diagram before moving on.

Message:

- Each layer separates a different responsibility.
- The separation is for diagnostics, caching, trust, and reproducibility.
- The point is not a generic compiler diagram. Each boundary says who owns a fact, which artifact records it, and what must be recomputed when it changes.

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3.2 - Responsibility Split (1/2)

Layer	Responsibility
frontend	source, lexing, parsing, recovery
resolver	imports, names, labels, namespaces
checker	soft types, clusters, registrations, overloads
elaborator	core logical representation
VC generator	proof and algorithm obligations

Layer	Row variables
Frontend	buttons, inputs, panels, outputs
Backend	DB queries, schemas, tasks, computations
Services	calls, queries, requests, notifications, commands
APIs/Protocols	data types, representations
IPC/Networking	protocols and algorithms, optimizations

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Responsibility Split (1/2). Point to the visible example, question, or diagram before moving on.

3.2 - Responsibility Split (2/2)

Layer	Responsibility
ATP	search for evidence
kernel	acceptance by replay/checking
artifact emitter	stable outputs for tools and dependents

Layer	Design rationale
UI/P	Search for evidence
Domain	Accountancy to replace checking
Business entities	Break out parts for tools and deployment

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Responsibility Split (2/2). Point to the visible example, question, or diagram before moving on.

Mizar-side semantics

-> well-typed, resolved obligations

ATP-side search

-> candidate proof evidence

kernel-side checking

-> accepted or rejected proof status

Key point:

- ATPs do not resolve names, infer types, expand clusters, or decide overloads.
- The kernel does not search for proofs.
- Deterministic pre-ATP discharge also needs replayable evidence or an explicit policy status; it is not accepted just because an earlier phase said "done".

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Mizar- $\omega$  side semantics
  -> not all applied, resolved obligations
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Design Reason:

- Putting all reasoning inside Mizar would make the trusted verifier larger.
- Delegating all reasoning to ATPs would lose control of Mizar-specific semantics.
- The hybrid boundary keeps semantic processing deterministic, ATP search powerful, and final acceptance independently checkable.

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Status distinction:

Status	Meaning
<code>kernel_verified</code>	replayed or certificate-checked evidence accepted
<code>discharged_builtin</code>	deterministic discharge accepted by the same evidence discipline
<code>externally_attested</code>	recorded backend or policy attestation, not equivalent to verified
<code>open / policy status</code>	unfinished or explicitly policy-controlled, not silently promoted

status distinction:	
status	Missing
overdue_perished	removed or considered rejected unless accepted
discharged_status	discharge accepted by the same evidence discipline
not accepted_status	removed because of policy information, not applicable: re-accepted
open / policy status	removed or rejected policy information, not discarded

Presenter script: For the table, read the row labels first, then the design reason. Use this as the transition: Reasoning Boundary (3/3). Point to the visible example, question, or diagram before moving on.

Explanation:

- High-level proof search may use ATPs.
- Accepted evidence is normalized into certificate data.
- The kernel checks imported facts, substitutions, alpha-conversion, clause well-formedness, and resolution/SAT evidence.
- Unsoundness in search should not become unsoundness in accepted results.
- The SAT part is proof evidence checking, not trust in a solver's success bit. Backend-specific proofs must be translated into replayable certificate data before acceptance.

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Certificate

```
target VC fingerprint
kernel profile
imported facts and hashes
generated clauses
substitutions
resolution trace
final goal
```

Message:

- The certificate binds a proof to a specific obligation and dependency slice.
- It must be deterministic and replayable.
- This is why the certificate carries hashes and a kernel profile: a proof accepted under one dependency and policy context should not silently migrate to another.

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After the example, say which design obligation the syntax makes visible.

Key Points:

- no proof search;
- no premise selection;
- no name resolution;
- no overload resolution;
- no cluster expansion;

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- no hidden global state;
- no trust in raw ATP success.

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Message:

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- Evo stores registration and cluster data in indexes.
- An active module gets an import-filtered view.
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resident memory should scale with:

- active source
- imported public interfaces
- import-filtered indexes
- active module VCs

not with:

- imported proof bodies
- private lemmas outside the interface
- unused registration data outside the import closure


```
res ident memory should scale with:  
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imported public interfaces  
import-filtered indexes  
acc ovr module Vfs  
  
not which:  
imported private bodies  
private lemma outside the interface  
unused registration data outside the import closure
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Memory Model. Point to the visible example, question, or diagram before moving on.

Main points:

- Hash source, dependency interfaces, generated VCs, and proof witnesses.
- Reuse only when fingerprints and verifier policy match.
- Proof-body-only changes should not rebuild unrelated importers when public statements and statuses remain unchanged.

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Main points:

- Parallelize independent modules, obligations, ATP runs, and kernel checks.
- Publish diagnostics, artifacts, and proof statuses in canonical order.
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Artifacts should serve:

- downstream package verification;
- IDE hover, diagnostics, and go-to-definition;
- AI context extraction and patch verification;
- documentation generation;
- Formalized Mathematics article links;

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- downstream package verification;
- IDE hover, diagnostics, and go-to-definition;
- AI context extraction and patch verification;
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- Formalized Mathematics article links;

Message:

- LSP gives editor feedback from syntax and semantic artifacts.
- Planned MCP-style resources and tools expose bounded verifier context to AI agents.
- Both should consume artifacts rather than reimplementing the verifier.
- The wire protocol is not the design center here; the design center is that agents read bounded, typed, auditable context and cannot bypass verification.
- The current fixed policy is the Green/Yellow/Red edit classification and authorization scopes; detailed wire schemas remain roadmap material.

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- Both should consume artifacts rather than reimplementing the verifier.
- The wire protocol is not the design center here; the design center is that agents read bounded, typed, auditable context and cannot bypass verification.
- The current fixed policy is the Green/Yellow/Red edit classification and authorization scopes; detailed wire schemas remain roadmap material.

Presenter script: Read aloud:

- LSP gives editor feedback from syntax and semantic artifacts.
- Planned MCP-style resources and tools expose bounded verifier context to AI agents.
- Both should consume artifacts rather than reimplementing the verifier.
- The wire protocol is not the design center here; the design center is that agents read bounded, typed, auditable context and cannot bypass verification.
- The current fixed policy is the Green/Yellow/Red edit classification and authorization scopes; detailed wire schemas remain roadmap material.

3.13 - Safe AI Edit Classes

Class	Examples	Policy
Green	add citation, insert qua, add info annotation	may be auto-proposed, still verified
Yellow	add import, local lemma, registration, invariant	proposed with human review
Red	add axiom, weaken theorem, change definition, relax kernel	forbidden to ordinary AI agents

Class	Examples	Notes
Class	add classes, remove class, add new annotations	may be auto-generated, not modified
Value	add inputs, local names, registers, sources	generated with some restrictions
Def	add axioms, make theorem, change definitions, reduce kernel	not done in standard AI agents

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Safe AI Edit Classes. Point to the visible example, question, or diagram before moving on.

Main principle:

```
Reject what must not pass  
before  
Accept everything that should pass
```

Talk angle:

- Parser gaps are annoying.
- Soundness bugs are dangerous.
- Kernel-adjacent tests should emphasize malformed and failing inputs.
- The test mix should therefore be intentionally asymmetric near the trust boundary: reject invalid evidence first, then broaden accepted-language coverage.

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Talk angle:

- Parser gaps are annoying.
- Soundness bugs are dangerous.
- Kernel-adjacent tests should emphasize malformed and failing inputs.
- The test mix should therefore be intentionally asymmetric near the test boundary: reject invalid evidence first, then broader accepted-but-gauge row bugs.

Presenter script: Read aloud:

- Parser gaps are annoying.
- Soundness bugs are dangerous.
- Kernel-adjacent tests should emphasize malformed and failing inputs.

After the example, say which design obligation the syntax makes visible.

Questions:

- Is the SAT certificate story convincing for Mizar-style proofs?
- Which proof evidence format would be easiest for the Mizar team to audit?
- Which artifact should be built first for real migration experiments?

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Presenter script: Read aloud:

- Is the SAT certificate story convincing for Mizar-style proofs?
- Which proof evidence format would be easiest for the Mizar team to audit?
- Which artifact should be built first for real migration experiments?

Key Phrase:

MML migration is a research program, not a one-shot translation.

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Roadmap Thesis. Point to the visible example, question, or diagram before moving on.

Outputs from the visit:

- a prioritized list of migration benchmark articles;
- agreement on compatibility metadata needs;
- review notes on language changes;
- a first paper outline;
- a shared list of objections and risks.

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Presenter script: Read aloud:

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- agreement on compatibility metadata needs;
- review notes on language changes;
- a first paper outline;
- a shared list of objections and risks.

Possible alpha scope:

- frontend and parser for a meaningful core subset;
- import prelude and module resolution prototype;
- structured AST and diagnostics;
- basic type and registration scaffolding;
- test harness and corpus layout;

Possible alpha scope:

- early artifact format for parsed/resolved source.

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Non-goals:

- full MML verification;
- final compatibility layer;
- stable AI protocol;
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- full MML verification;
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Presenter script: Read aloud:

- full MML verification;
- final compatibility layer;
- stable AI protocol;
- complete Formalized Mathematics workflow.

Focus:

- choose 3-5 representative MML articles;
- translate by hand and with scripts;
- record every mismatch as a classified issue;
- build side-by-side reports.

Suggested issue classes:

- syntax translation;
- environment-to-import mapping;
- structure inheritance;
- registration or cluster behavior;
- overload behavior;

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- translate by hand and with scripts;
- record every mismatch as a classified issue;
- build side-by-side reports.

Suggested issue classes:

- syntax translation;
- sentence-to-sentence mapping;
- structural distance;
- regulations or climate behavior;
- overall behavior;

Presenter script: Read aloud:

- choose 3-5 representative MML articles;
- translate by hand and with scripts;
- record every mismatch as a classified issue;
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- syntax translation;

Suggested issue classes:

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Presenter script: Read aloud:

- proof citation or ATP behavior;
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Candidate path:

- ① foundational set and relation fragments;
- ② functions and binary operations;
- ③ algebraic structures;
- ④ selected theorem-heavy articles;
- ⑤ broader dependency cones around successful fragments.

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Measure:

- translated LOC and article count;
- accepted parser subset;
- resolved imports versus unresolved dependencies;
- proof obligations generated;
- obligations closed by deterministic reasoning;

Measure:

- obligations closed by ATP plus kernel certificate;
- memory and build time per module;
- number of compatibility decisions needing human review.

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Presenter script: Read aloud:

- translated LOC and article count;
- accepted parser subset;
- resolved imports versus unresolved dependencies;
- proof obligations generated;
- obligations closed by deterministic reasoning;

Possible policy:

- preserve theorem identity through origin metadata;
- keep source-visible compatibility aliases where they help migration;
- avoid preserving legacy forms that block clear module and artifact semantics;
- record every divergence from old behavior with a reason and test.
- Compatibility should be argued from mathematical identity, review value, and reproducibility, not from nostalgia for every surface form.

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4.8 - Migration Risk Table

Risk	Mitigation
syntax compatibility consumes the project	start with representative slices, not full automation
registrations behave differently	build trace and explanation artifacts early
proof search differs from current verifier	record used facts and compare proof obligations
package layout breaks Formalized Mathematics links	origin metadata and article-to-library identifiers
AI edits hide migration mistakes	keep Red edits forbidden and require verifier artifacts

└ Part 4. Roadmap

└ 4.8 - Migration Risk Table

4.8 - Migration Risk Table

Item	Migration
quantify compatibility between properties	start with representative classes, not full automation
regeneration behavior of classes	use ML trace on deep learning artifacts only
proof search algorithm current version	extend used facts and compare proof obligations
package types: binary form of the knowledge	single modules and modules - binary knowledge
Notes	
All risk table migration migration	keep the data in the table and require further analysis

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Migration Risk Table. Point to the visible example, question, or diagram before moving on.

Questions:

- Which MML articles are small but structurally representative?
- Which article families stress registrations and structures?
- Which current Mizar idioms are culturally important, not just technically convenient?
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- Which MML articles are small but structurally representative?
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- What migration result would convince the community that Evo is serious?

Key Phrase:

The verified library and the published article should be linked but not forced to have the same structure.

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Publication Thesis. Point to the visible example, question, or diagram before moving on.

Library modules optimize:

- reuse;
- dependency control;
- package versioning;
- stable machine-readable artifacts.

Formalized Mathematics articles optimize:

- exposition;
- narrative;
- citation;
- peer review;
- reader-facing mathematical structure.

Library modules optimize:

- reuse;
- dependency control;
- package versioning;
- stable machine-readable artifacts.

Formalized Mathematics articles optimize:

- exposition;
- notation;
- citation;
- peer review;
- reading/finding mathematical literature.

Presenter script: Read aloud:

- reuse;
- dependency control;
- package versioning;
- stable machine-readable artifacts.
- exposition;

Formalized Mathematics theorem

- > library package
- > module path
- > theorem FQN
- > statement fingerprint
- > verified artifact hash
- > rendered documentation page

```
Formalized Mathematics theorems
-> library package
-> module path
-> theorems TTH
-> statement fingerprint
-> verified statement hash
-> rendered documentation page
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Link Model. Point to the visible example, question, or diagram before moving on.

5.4 - Example Link Record (1/2)

```
article: Formalized Mathematics, future volume
section: Unique identity element
library_object: std.algebra.group.identity_unique
package: std_algebra
version: 0.4.0
statement_hash: sha256:...
artifact_hash: sha256:...
origin: mizar:GROUP_1:...
```

```
name: Formalized Mathematics, future volume  
description: the type identity elements  
library_subject: std.algebra.group, identity, unique  
package: std_algebra  
version: 0.0.0  
creation_date: 2023-06-18  
last_modified: 2023-06-18  
origin: mizar:000000_01...
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Example Link Record (1/2). Point to the visible example, question, or diagram before moving on.

Design Reason:

- These identities are deliberately layered.
- The article label supports scholarly citation and narrative structure.
- The origin id preserves continuity with MML and existing Formalized Mathematics history.
- The library FQN addresses the current reusable theorem object.
- The statement fingerprint detects semantic drift.

Design Reason:

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For readers:

- article prose remains primary;
- formal source is one click away;
- proof status and dependency information are inspectable.

For maintainers:

- library refactoring does not automatically rewrite article exposition;
- versioned links preserve reproducibility;
- origin metadata preserves historical continuity.

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For AI:

- article prose provides semantic search text;
- formal links provide exact theorem identities;
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- agents can retrieve both explanation and machine-checkable context.

Questions:

- Should stable identity be based on library FQN, article label, origin id, or artifact hash?
- Which of these identities should be primary in user-facing citations, and which should remain machine-facing reproducibility checks?
- How should article revisions track library refactoring?
- Should Formalized Mathematics publish generated HTML, PDF, or both?
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5.7 - Possible Workflow

library theorem verifies

- > verified artifact emitted
- > documentation generated
- > article references formal objects
- > publication build checks links and versions
- > article published with formal links

```
library has been verified
-> verified set of set emitted
-> document is generated
-> article references formal objects
-> publication build should take several years
-> article published with formal links
```

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Possible Workflow. Point to the visible example, question, or diagram before moving on.

Key Points:

- language identity and compatibility;
- MML migration examples;
- registration and cluster semantics;
- proof evidence and kernel boundary;
- Formalized Mathematics link model;

Key Points:

- roadmap realism.

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Presenter script: Read aloud:

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Proposed next steps:

- ➊ Select migration benchmark articles.
- ➋ Prepare exact old/new code comparison slides.
- ➌ Write an initial MML migration report template.
- ➍ Refine this detailed Beamer deck after Bialystok review.
- ➎ Start the paper outline using the same main claims and review feedback.

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- Prepare exact old/new code comparison slides.
- Write an initial MML migration report template.
- Refine this detailed Beamer deck after Bialystok review.
- Start the paper outline using the same main claims and review feedback.

Final slide:

Mizar Evo should be modern where scale demands it,
and conservative where Mizar's mathematical identity depends on it.

Presenter script: After the example, say which design obligation the syntax makes visible. Use this as the transition: Closing. Point to the visible example, question, or diagram before moving on.

Prepared short excerpts for Beamer conversion:

[Article environment](#) **Source:** algstr_0.miz

Lines: 15-25

Migration point: Translate article-level dependency classes into explicit import/package roles.

Evo sketch: `import std.function.binop; import std.algebra.structure.sorted;`

Prepared short excerpts for Beamer conversion:

[Structure definition](#) **Source:** algstr_0.miz

Lines: 37-40

Migration point: Preserve structures while separating intrinsic fields from canonical properties and inheritance obligations.

Evo sketch: `field carrier, field add, and, where appropriate, property zero /
property unit`

Mizar Evo

└ Backup A. Prepared Exact Examples

└ Backup A. Prepared Exact Examples (1/4)

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Backup A. Prepared Exact Examples (1/4). Point to the visible example, question, or diagram before moving on.

Prepared short excerpts for Beamer conversion:

Arbitrary example: Source: `sgen2_e.miz`

Lines: 15-25

Migration point: This does a variable-level dependency check into `import/package`

rules.

Key words: `import`, `end.function.hint`; `import`, `end.algebra.structure.assert`;

Prepared short excerpts for Beamer conversion:

Arbitrary example: Source: `sgen2_e.miz`

Lines: 37-48

Migration point: Prepare structures while separating intrinsic rules from canonical

prop as known inference subgoals.

Key words: `field`, `cancel`, `field`, `add`, `end`, where appropriate, `prophecy`, `area` /

`prophecy`, `unit`

Prepared short excerpts for Beamer conversion:

[Registration/cluster propagation](#) **Source:** algstr_0.miz

Lines: 104-109

Migration point: Keep registrations, but make propagation traceable and import-filtered.

Evo sketch: registration graph entry plus resolution trace artifact

Prepared short excerpts for Beamer conversion:

[Proof citation](#) **Source:** struct_0.miz

Lines: 637-643

Migration point: Treat citation repair as a small verifier-checked edit.

Evo sketch: Green edit: add or refine by citations; artifact records used facts

└ Backup A. Prepared Exact Examples

└ Backup A. Prepared Exact Examples (2/4)

Presenter script: For the table, read the row labels first, then the design reason.

Use this as the transition: Backup A. Prepared Exact Examples (2/4). Point to the visible example, question, or diagram before moving on.

Prepared short excerpts for Beamer conversion:

Register/trace propagation Sources: register_e.mex

Issue: 1164-118

Migration point: Keep register/trace, but make propagation traceable and in post-thread.

Key sketch: registration graph entry plus random trace artifact

Prepared short excerpts for Beamer conversion:

Final sketch Sources: register_e.mex

Issue: 1164-118

Migration point: Trace/traceable repair as a small non-checked edit.

Key sketch: Green edit: add or refine by disjoining artifact records used later

Prepared short excerpts for Beamer conversion:

[Publication link](#) **Source:** `contents.html`

Lines: 516-517

Migration point: Link scholarly article metadata to a reusable MML/library identifier.

Evo sketch: Formalized Mathematics article label + origin: `mizar:GROUP_1 + theorem FQN/fingerprint`

Source URLs:

- ALGSTR_0: `<https://mizar.uwb.edu.pl/version/current/mml/algstr_0.miz>`
- STRUCT_0: `<https://mizar.uwb.edu.pl/version/current/mml/struct_0.miz>`
- Formalized Mathematics contents: `<https://mizar.uwb.edu.pl/fm/contents.html>`

Mizar Evo

└ Backup A. Prepared Exact Examples

└ Backup A. Prepared Exact Examples (3/4)

Prepared short excerpts for Beamer conversion:

Published in: Source: c.sciences.uwb.edu.pl

Issue: 2026-06-18

Migration path: Use manually available metadata to a suitable MML/Export identifier.

File sketch: Formalized Mathematics content @ original: mizar:current_c@

chosen: 00.00.00.00.00

Source URLs:

- ALGSTR_0: <https://mizar.uwb.edu.pl/version/current/mml/algstr_0.miz>
- STRUCT_0: <https://mizar.uwb.edu.pl/version/current/mml/struct_0.miz>
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Presenter script: Read aloud:

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Remaining Beamer preparation:

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Required diagrams:

- 1 Current Mizar workflow: source -> Accommodator -> Verifier -> Exporter -> MML.
- 2 Three pillars: readability, AI-readiness, scalability.
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- 4 Structure inheritance and diamond coherence.
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Required diagrams:

- ① Reasoning boundary: Mizar semantics / ATP search / kernel checking.
- ② Certificate object and SAT/UNSAT checking.
- ③ Incremental dependency and fingerprint graph.
- ④ AI patch verification flow.
- ⑤ Formalized Mathematics article-to-library link model.

Required diagrams:

- ① Roadmap timeline.

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Possible paper title:

Mizar Evo: Readable, AI-Ready, and Scalable Formal Mathematics

Possible sections:

- ➊ Introduction: why Mizar needs evolution now.
- ➋ Mizar as baseline: readability, MML, Formalized Mathematics.
- ➌ Design principles.
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- ➎ Verifier architecture and small kernel.

Mizar Evo

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After the example, say which design obligation the syntax makes visible.

Possible sections:

- 1 AI-safe proof development.
- 2 Package-based library and publication workflow.
- 3 Migration plan and evaluation metrics.
- 4 Related work: current Mizar, Lean, Isabelle/Isar, ATP-integrated systems.
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Mizar Evo

└ Backup C. Paper Outline Seed

└ Backup C. Paper Outline Seed (2/2)

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Use this checklist before converting to Beamer:

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- Does every Lean comparison avoid sounding dismissive?
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- Are MML migration claims measurable?
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